Galois theory

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Abstract. The notes correspond to the bachelor course **Galois Theory** of the Vrije Universiteit Brussel, Faculty of Sciences, Department of Mathematics and Data Sciences.

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Contents

Introduction

The notes correspond to the bachelor course **Galois theory** of the Vrije Universiteit Brussel, Faculty of Sciences, Department of Mathematics and Data Science. The course is divided into twelve two-hour lectures.

The material is somewhat standard. Basic texts on fields and Galois theory are for example [4] and [5].

As usual, we also mention a set of great expository papers by Keith Conrad. The notes are extremely well-written and useful at every stage of a mathematical career.

The notes include many exercises, some with detailed solutions. Mandatory exercises have a green background, while optional ones (bonus exercises) have a yellow background. The notes also include some additional comments. While these are entirely optional, I hope they offer further insight. They are highlighted with a pink background.

The notes include Magma code, which we use to verify examples and offer alternative solutions to certain exercises. Magma [1] is a powerful software tool designed for working with algebraic structures. There is a free online version of Magma available.

Several of the exercise solutions were written by Silvia Properzi.

These notes are freely available at `https://github.com/vendramin/galois`, where the most recent version can always be found. They are also listed in the AMS Open Math Notes collection, reference `OMN:202508.111468`; the copy hosted there is a snapshot and may lag behind the version on GitHub.

If you want to cite these notes, you may use the following entry:

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§ 1. Fields

Lecture 1

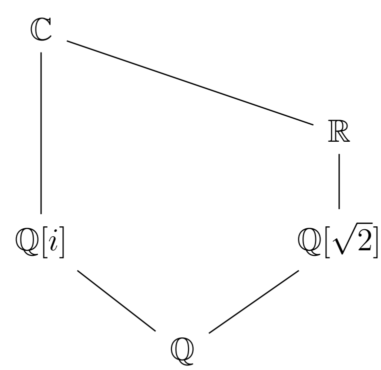
Throughout these notes, rings have an identity. Ring homomorphisms preserve the identity, and subrings contain the identity of the ambient ring.

Recall that a **field** is a commutative ring such that $1 \neq 0$ and every non-zero element is invertible. Examples of (infinite) fields are $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$. If p is a prime number, then $\mathbb{Z}/p$ is a field.

1.1. Example. The abelian group $\mathbb{Z}/2 \times \mathbb{Z}/2$ is a field with multiplication

$$(a,b)(c,d) = (ac+bd, ad+bc+bd).$$

1.2. Example. $\mathbb{Q}[i] = \{a+bi : a,b \in \mathbb{Q}\}$ and $\mathbb{Q}[\sqrt{2}] = \{a+b\sqrt{2} : a,b \in \mathbb{Q}\}$ are fields.



1.3. Exercise. Prove that $\mathbb{Q}[i]$ and $\mathbb{Q}[\sqrt{2}]$ are not isomorphic as fields.

If R is a ring, there exists a unique ring homomorphism $\mathbb{Z} \to R$, $m \mapsto m1$. The image

$$R_{\mathbb{Z}} = \{m1 : m \in \mathbb{Z}\}$$

of this homomorphism is a subring of R, and it is known as the **prime subring** of R. The kernel is a subgroup of $\mathbb{Z}$ generated by some $t \geq 0$. The integer t is the **characteristic** of the ring R.

1.4. Exercise. The characteristic of a field is either zero or a prime number.

1.5. Example. The characteristic of the field of Example 1.1 is two. Why?

Recall that a non-zero commutative ring R is an **integral domain** if $xy = 0 \implies x = 0$ or $y = 0$. Fields are integral domains.

1.6. Exercise. Let K be a field. Prove that the following statements are equivalent:

1) K is of characteristic zero.
2) The additive order of 1 is infinite.
3) The additive order of each $x \neq 0$ is infinite.
4) The prime subring of K is isomorphic to $\mathbb{Z}$.

1.7. Exercise. Let p be a prime number and let K be a field. Prove that the following statements are equivalent:

1) K is of characteristic p.
2) The additive order of 1 is p.
3) The additive order of each $x \neq 0$ is p.
4) The prime subring of K is isomorphic to $\mathbb{Z}/p$.

Note that finite fields are of prime characteristic.

1.8. Lemma. *Let R be a commutative ring, $z \in R$ and $f(X) \in R[X]$ be a non-constant polynomial. Then z is a root of f if and only if $X - z$ divides f in $R[X]$.*

In particular, if R is an integral domain, then f has at most $\deg(f)$ *roots in R.*

Proof. Clearly, if $X - z$ divides f in $R[X]$, then z is a root of f.

Let $n = \deg(f)$ and $f = \sum_{i=0}^{n} a_i X^i$, for some $a_0, \dots, a_n \in R$. Then

$$f(X) - f(z) = \sum_{i=0}^{n} a_i X^i - \sum_{i=0}^{n} a_i z^i = \sum_{i=0}^{n} a_i (X^i - z^i) = \sum_{i=1}^{n} a_i (X^i - z^i).$$

Note that for every integer $i > 0$,

$$X^i - z^i = \sum_{k=0}^{i-1} z^k X^{i-k} - \sum_{k=0}^{i-1} z^{k+1} X^{i-1-k} = (X - z) \sum_{k=0}^{i-1} z^k X^{i-k-1},$$

hence $X - z$ divides $X^i - z^i$ in $R[X]$ for every $i > 0$. So $X - z$ divides $f(X) - f(z)$ in $R[X]$. If we now assume that z is a root of f, then $f(X) = f(X) - f(z)$ and it is divisible by $X - z$ in $R[X]$.

Suppose now that R is an integral domain. We prove the second part of the lemma by induction on the degree of f. If f has degree 1, then $f(X) = aX + b$ for some $a, b \in R$ with $a \neq 0$. Given two roots $z_1, z_2 \in R$ of f,

$$az_1 = -b = az_2,$$

so $a(z_1 - z_2) = 0$. But since R is a domain and $a \neq 0$, necessarily $z_1 = z_2$. Assume now that $\deg(f) = n > 1$ and that every non-constant polynomial $g(X) \in R[X]$ of degree $< n$ has at most $\deg(g)$ roots in R. If f has no roots in R, then the result is immediate. If f has a root $z \in R$, then, by the previous part of this lemma, there exists $g(X) \in R[X]$ such that

$$f(X) = (X - z)g(X).$$

Hence, $0 < \deg(g) = \deg(f) - 1 = n - 1$. Therefore, we can apply the inductive hypothesis to g and obtain that g has at most $n - 1$ roots in R. To conclude, we can observe that $t \in R$ is a root of f if and only if $0 = f(t) = (t - z)g(t)$. Since R is a domain, the last equality is equivalent to $t = z$ or $g(t) = 0$. Thus f has at most n distinct roots. □

Note that for the second part of the previous lemma, the hypothesis that R is an integral domain is necessary, as the following examples show.

1.9. Example. The polynomial $X^2 - 1 \in (\mathbb{Z}/8)[X]$ has 4 roots in $\mathbb{Z}/8$: 1,3,5, and 7.

1.10. Example. Let $R = H(\mathbb{R})$ be the ring of real quaternions. Then $X^2 + 1 \in R[X]$ has infinitely many roots in R.

1.11. PROPOSITION. *Let K be a field. Then any finite subgroup of $K^\times = K \setminus \{0\}$ is cyclic.*

PROOF. Let G be a finite subgroup of $K^\times$. Let $x \in G$ have maximal order N. We will show that any element of G is a power of x. Let $y \in G$ and $n = |y|$.

We claim that n divides N. If not, there exists a prime number p and a power $q = p^{\alpha+1}$ of p such that $p^{\alpha+1} \mid n$ and $p^\alpha \mid N$, but $p^{\alpha+1} \nmid N$. Let $z = x^{p^\alpha} y^{n/q}$. Then $|x^{p^\alpha}| = N/p^\alpha$ and $|y^{n/q}| = q$. Since $\gcd(N/p^\alpha, q) = 1$ and G is abelian,

$$|z| = \operatorname{lcm}\{N, q\} = Np > N,$$

a contradiction.

Note that $X^n - 1 \in K[X]$ has at most n roots in K. Moreover, $x^{kN/n}$ for $k \in \{0, \dots, n-1\}$ are n distinct elements of K, that are roots of $X^n - 1$. Therefore

$$\{x^{kN/n} \mid k \in \{0, \dots, n-1\}\}$$

is the set of all roots of $X^n - 1 \in K[X]$ in K. Since y has order n, it has to be one of these roots. Thus $y = x^{kN/n}$ for some $k \in \{0, \dots, n-1\}$. □

1.12. DEFINITION. A **subfield** of a ring R is a subring of R that is also a field.

Note that if K is a subfield of E, then the characteristic of K coincides with the characteristic of E. Moreover, if $K \to L$ is a field homomorphism, then K and L have the same characteristic.

1.13. EXERCISE. Let K be a field of characteristic $p > 0$. Prove that $K \to K$, $x \mapsto x^{p^n}$, is a field homomorphism for all $n \in \mathbb{Z}_{\geq 0}$.

1.14. DEFINITION. A field K is **prime** if there are no proper subfields of K.

Examples of prime fields are $\mathbb{Q}$ and $\mathbb{Z}/p$ for a prime number p.

1.15. PROPOSITION. *Let K be a field. The following statements hold:*

1) *K has a unique subfield which is prime; it is known as the* ***prime subfield*** *of K.*
2) *The prime subfield of K is either isomorphic to $\mathbb{Q}$ if the characteristic of K is zero, or it is isomorphic to $\mathbb{Z}/p$ for some prime number p if the characteristic of K is p.*

PROOF. To prove the first claim let L be the intersection of all the subfields of K. Then L is a subfield of K. If F is a subfield of L, then F is a subfield of K. Thus $L \subseteq F$ and hence $F = L$, which proves that L is prime. If L_1 is a subfield of K and L_1 is prime, then $L \subseteq L_1$ and hence $L = L_1$.

Let K_0 be the prime subfield of K. Suppose that K is of characteristic $p > 0$. Then the prime subring $K_\mathbb{Z}$ of K is a field isomorphic to $\mathbb{Z}/p$ and hence $K_0 = K_\mathbb{Z} \simeq \mathbb{Z}/p$. Suppose now that the characteristic of K is zero. Let $E = \{m1(n1)^{-1} : m, n \in \mathbb{Z}, n \neq 0\}$. The map $\mathbb{Q} \to E$, $m/n \mapsto m1(n1)^{-1}$, is a field isomorphism. We claim that $K_0 = E$. Since $K_\mathbb{Z} \subseteq K_0$, it follows that $E \subseteq K_0$. Hence $E = K_0$, as E is a subfield of K. □

1.16. DEFINITION. Let E be a field and K be a subfield of E. Then E is a **field extension** of K. We will use the notation E/K.

Let E/K be a field extension. Then E is a K-vector space with the usual scalar multiplication $K \times E \to E$, $(\lambda, x) \mapsto \lambda x$.

1.17. Definition. The **degree** of an extension E of K is the dimension $\dim_K E$, which may be infinite. It will be denoted by $[E : K]$.

We say that E is a **finite extension** of K if $[E : K]$ is finite.

1.18. Example. Let K be a field. Then $[K : K] = 1$. Conversely, if E is an extension of K and $[E : K] = 1$, then $K = E$. If not, let $x \in E \setminus K$. We claim that $\{1, x\}$ is linearly independent over K. Indeed, if $a1 + bx = 0$ for some $a, b \in K$, then $bx = -a$. If $b \neq 0$, then $x = -a/b \in K$, a contradiction. If $b = 0$, then $a = 0$.

We know that $[\mathbb{C} : \mathbb{R}] = 2$.

1.19. Example. A basis of $\mathbb{Q}[\sqrt{2}]$ over $\mathbb{Q}$ is given by $\{1, \sqrt{2}\}$. Then $[\mathbb{Q}[\sqrt{2}] : \mathbb{Q}] = 2$. The calculations can be easily done with Magma:

```
> Q<z> := QuadraticField(2);
> Characteristic(Q);
0
> K := PrimeField(Q);
> K;
Rational Field
> Degree(Q);
2
> Basis(Q);
[ 1, z ]
> z^2;
2
> One(K) eq One(Q);
true
> G := GaloisGroup(Q);
> G;
Symmetric group G acting on a set of cardinality 2
Order = 2
```

1.20. Example. Since $\mathbb{Q}$ is countable and $\mathbb{R}$ is not, $[\mathbb{R} : \mathbb{Q}] > \aleph_0$. If $\{x_i : i \in \mathbb{Z}_{>0}\}$ is a countable basis of $\mathbb{R}$ over $\mathbb{Q}$, for each n consider the $\mathbb{Q}$-vector space V_n generated by $\{x_1, \dots, x_n\}$. Each V_n is countable, hence

$$\mathbb{R} = \bigcup_{n \geq 1} V_n$$

is countable, a contradiction.

If E is an extension of K and E is finite as a set, then $[E : K]$ is finite.

1.21. Proposition. *Let K be a finite field. Then $|K| = p^m$ for some prime number p and some $m \geq 1$.*

Proof. We know the prime subfield K_0 of K is isomorphic to $\mathbb{Z}/p$. In particular, $|K_0| = p$. Since K is finite, $[K : K_0] = m$ for some m. If $\{x_1, \dots, x_m\}$ is a basis of K over K_0, then each element of K can be written uniquely as $\sum_{i=1}^{m} a_i x_i$ for some $a_1, \dots, a_m \in K_0$. Then there is a bijection between K and K_0^m and hence $|K| = |K_0^m| = p^m$. $\square$

We now perform some basic calculations with a finite field of eight elements:

```
> E<x> := FiniteField(8);
> PrimeField(E);
Finite field of size 2
> Degree(E);
3
> Characteristic(E);
2
> #E;
8
> { x : x in E };
{ 1, x, x^2, x^3, x^4, x^5, x^6, 0 }
```

Here is an alternative construction of a field of eight elements as a quotient of a polynomial ring:

```
> R<x> := PolynomialRing(PrimeField(E));
> IsIrreducible(x^3+x+1);
true
> I := ideal< R | x^3+x+1 >;
> IsMaximal(I);
true
> Q<y> := quo< R | I >;
> IsField(Q);
true
> { y : y in Q };
{
    y^2 + 1,
    0,
    y^2,
    y + 1,
    1,
    y,
    y^2 + y + 1,
    y^2 + y
}
> #Q;
8
```

1.22. Definition. Let E be an extension of K. A **subextension** F/K of E/K is a subfield F of E that contains K, that is $K \subseteq F \subseteq E$.

1.23. Definition. Let E and E_1 be extensions over K. An **extension homomorphism**

$$E/K \to E_1/K$$

is a field homomorphism $\sigma\colon E \to E_1$ such that $\sigma(x) = x$ for all $x \in K$.

To describe the homomorphism $\sigma\colon E/K \to E_1/K$ of the extensions over K one typically writes the commutative diagram

$$\begin{tikzcd} K \arrow[r, equal] \arrow[d, hook] & K \arrow[d, hook] \\ E \arrow[r, "\sigma"] & E_1 \end{tikzcd}$$

We write $\mathrm{Hom}(E/K, E_1/K)$ to denote the set of homomorphisms $E/K \to E_1/K$ of extensions of K. Note that if $\sigma \in \mathrm{Hom}(E/K, E_1/K)$, then σ is a K-linear map, as

$$\sigma(\lambda x) = \sigma(\lambda)\sigma(x) = \lambda\sigma(x)$$

for all $\lambda \in K$ and $x \in E$.

1.24. Example. The conjugation map $\mathbb{C} \to \mathbb{C}$, $z \mapsto \overline{z}$, is an endomorphism of $\mathbb{C}$ as an extension over $\mathbb{R}$. Let $\varphi \in \mathrm{Hom}(\mathbb{C}/\mathbb{R}, \mathbb{C}/\mathbb{R})$. Then

$$\varphi(x+iy) = \varphi(x) + \varphi(i)\varphi(y) = x + \varphi(i)y$$

for all $x, y \in \mathbb{R}$. Since $\varphi(i)^2 = \varphi(i^2) = \varphi(-1) = -1$, it follows that $\varphi(i) \in \{-i, i\}$. Thus either $\varphi(x+iy) = x+iy$ or $\varphi(x+iy) = x-iy$.

1.25. Exercise. Let K be a field, K_0 be its prime subfield and $\sigma\colon K \to K$ be a field homomorphism. Prove that $\sigma \in \mathrm{Hom}(K/K_0, K/K_0)$.

If E/K is an extension, then

$$\begin{aligned} \mathrm{Aut}(E/K) &= \{\sigma\colon E/K \to E/K \text{ is a bijective extension homomorphism}\} \\ &= \{\sigma\colon E \to E : \sigma \text{ is a bijective field homomorphism with } \sigma|_K = \mathrm{id}\} \end{aligned}$$

is a group with composition.

1.26. Definition. Let E/K be an extension. The **Galois group** of E/K is the group $\mathrm{Aut}(E/K)$ and it will be denoted by $\mathrm{Gal}(E/K)$.

A typical example: $\mathrm{Gal}(\mathbb{C}/\mathbb{R}) \simeq \mathbb{Z}/2$.

As an example, we show with the computer that $\mathrm{Gal}(\mathbb{Q}[\sqrt{2}]/\mathbb{Q}) \simeq \mathbb{Z}/2$:

```
> Q<z> := QuadraticField(2);
> G := GaloisGroup(Q);
> G;
Symmetric group G acting on a set of cardinality 2
Order = 2
> GroupName(G);
C2
```

1.27. Exercise. Prove that the polynomial $X^3 - 2$ is irreducible in $\mathbb{Q}[X]$.

The previous exercise can easily be solved using Magma:

```
> Q<x> := PolynomialRing(Rationals());
> f := x^3-2;
> IsIrreducible(f);
true
```

1.28. Example. Let $\theta = \sqrt[3]{2}$ and let $E = \{a + b\theta + c\theta^2 : a, b, c \in \mathbb{Q}\}$. Note that

$$a + b\theta + c\theta^2 = 0 \Longleftrightarrow a = b = c = 0.$$

Since $X^3 - 2$ is irreducible in $\mathbb{Q}[X]$, evaluation at θ yields $E \simeq \mathbb{Q}[X]/(X^3 - 2)$. Then E is an extension of $\mathbb{Q}$ such that $[E : \mathbb{Q}] = 3$. We claim that $\mathrm{Gal}(E/\mathbb{Q})$ is trivial. If $\sigma \in \mathrm{Gal}(E/\mathbb{Q})$ and $z = a + b\theta + c\theta^2$, then $\sigma(z) = a + b\sigma(\theta) + c\sigma(\theta)^2$. Since $\sigma(\theta) \in E \subseteq \mathbb{R}$ and

$$\sigma(\theta)^3 = \sigma(\theta^3) = \sigma(2) = 2,$$

it follows that $\sigma(\theta) = \theta$ and therefore $\sigma = \mathrm{id}$.

Recall that we have the following characterization of irreducible polynomials, known as **Gauss' Lemma**.

1.29. Lemma (Gauss' lemma). *Let A be a unique factorization domain and K be its fraction field. A non-constant polynomial $f \in A[X]$ is irreducible in $A[X]$ if and only if it is primitive and irreducible in $K[X]$.*

In particular, a non-constant polynomial $f \in \mathbb{Z}[X]$ is irreducible if and only if it is primitive and irreducible in $\mathbb{Q}[X]$. This helps with the following exercise, which is known as **Eisenstein's irreducibility criterion**:

1.30. Exercise. Let $f = \sum_{i=0}^{n} a_i X^i \in \mathbb{Z}[X]$ be a polynomial of degree $n > 0$. Assume that there exists a prime number p such that $p \mid a_i$ for all $i \in \{0, 1, \dots, n-1\}$, $p \nmid a_n$ and $p^2 \nmid a_0$. Then f is irreducible in $\mathbb{Q}[X]$.

1.31. Bonus exercise. Let A be a unique factorization domain and K be its fraction field. Let $f = \sum_{i=0}^{n} a_i X^i \in A[X]$ be a polynomial of degree $n > 0$. Assume that there exists a prime element $p \in A$ such that $p \mid a_i$ for all $i \in \{0, 1, \dots, n-1\}$, $p \nmid a_n$ and $p^2 \nmid a_0$. Then f is irreducible in $K[X]$.

1.32. Exercise. Prove that the polynomials

$$f = X^{10} + 60X^7 + 82X^6 - 36X^3 + 2,$$
$$g = 3X^{10} + 15X^2 - 45,$$

are irreducible in $\mathbb{Q}[X]$.

The previous exercise is easy for Magma:

```
> R<x> := PolynomialRing(Rationals());
> g := 3*x^10+15*x^2-45;
> IsIrreducible(g);
true
> f := x^10+60*x^7+82*x^6-36*x^3+2;
> IsIrreducible(f);
true
```

1.33. Exercise. Is the polynomial $f = 3(X^{10} + 5X^2 - 15)$ irreducible in $\mathbb{Z}[X]$?

Let us see what Magma can do here:

```
> R<x> := PolynomialRing(Integers());
> f := 3*x^10+15*x^2-45;
> IsIrreducible(f);
false
> Factorization(f);
[
    <3, 1>,
    <x^10 + 5*x^2 - 15, 1>
]
```

If E/K is an extension and S is a subset of E, then there exists a unique smallest subextension F/K of E/K such that $S \subseteq F$. In fact,

$$F = \bigcap \{T : T \text{ is a subfield of } E \text{ that contains } K \cup S\}.$$

If L/K is a subextension of E/K such that $S \subseteq L$, then $F \subseteq L$ by definition. The extension F is known as the **subextension generated by** S and it will be denoted by $K(S)$. If $S = \{x_1, \dots, x_n\}$ is finite, then the extension $K(S)/K$, where $K(S) = K(x_1, \dots, x_n)$, is said to be of **finite type**.

1.34. Example. If $\{e_1, \dots, e_n\}$ is a basis of E over K, then $E = K(e_1, \dots, e_n)$.

1.35. Example. The field $\mathbb{Q}(\sqrt{2})$ is precisely the subextension of $\mathbb{R}/\mathbb{Q}$ generated by $\sqrt{2}$.

Let E/K be an extension and S and T be subsets of E. Then

$$K(S \cup T) = K(S)(T) = K(T)(S).$$

If, moreover, $S \subseteq T$, then $K(S) \subseteq K(T)$.

§ 2. Algebraic extensions

2.1. Definition. Let E/K be an extension. An element $x \in E$ is **algebraic** over K if there exists a non-zero polynomial $f(X) \in K[X]$ such that $f(x) = 0$. If x is not algebraic over K, then it is called **transcendental** over K.

2.2. Definition. An extension E/K is **algebraic** if every $x \in E$ is algebraic over K.

If K is a field, every $x \in K$ is algebraic over K, as x is a root of $X - x \in K[X]$. In particular, K/K is an algebraic extension.

2.3. Example. $\mathbb{C}/\mathbb{R}$ is an algebraic extension. If $z \in \mathbb{C} \setminus \mathbb{R}$, then z is a root of the polynomial $X^2 - (z + \overline{z})X + |z|^2 \in \mathbb{R}[X]$.

If F/K is an extension and $x \in E$ is algebraic over K for some field $E \supseteq F$, then x is algebraic over F.

2.4. Example. $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is algebraic, as the number $a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$ is a root of the polynomial $X^2 - 2aX + (a^2 - 2b^2) \in \mathbb{Q}[X]$.

The extension $\mathbb{C}/\mathbb{Q}$ is not algebraic. For example, Hermite proved that e is transcendental over $\mathbb{Q}$; see [**5**, Theorem 24.4]. Lindemann's theorem states that π is not algebraic over $\mathbb{Q}$; see [**5**, Theorem 24.5].

2.5. Example. Let $a = \sqrt{2}$ and $b = \sqrt[3]{3}$. Both a and b are algebraic numbers over $\mathbb{Q}$. Let us show that $a+b$ is also algebraic. Let $f(X) = X^3 - 3 \in \mathbb{Q}[X]$. Then $f(b) = 0$. Note that the polynomial

$$g(X) = f(X-a) = X^3 - 3aX^2 + 3a^2X - a^3 - 3 \in \mathbb{Q}(a)[X]$$

is such that $g(a+b) = 0$. How can we find a polynomial with coefficients in $\mathbb{Q}$ that vanishes on $a+b$? We do the "conjugation" trick:

$$h(X) = f(X-a)f(X+a) = X^6 - 6X^4 - 6X^3 + 12X^2 - 36X + 1 \in \mathbb{Q}[X].$$

Note that $h(a+b) = 0$. How can you prove that ab is also algebraic over $\mathbb{Q}$?

If E/K is an extension and $x \in E$ is algebraic over K, then the evaluation homomorphism $K[X] \to E$, $p \mapsto p(x)$, is not injective. In particular, its kernel is a non-zero ideal. Hence it is generated by a monic polynomial f.

2.6. Definition. Let E/K be an extension and $x \in E$ be an algebraic element. The monic polynomial that generates the kernel of $K[X] \to E$, $f \mapsto f(x)$, is known as the **minimal polynomial** of x over K and it will be denoted by $f(x, K)$. The **degree** of x over K is then $\deg f(x, K)$.

Some basic properties of the minimal polynomial of an algebraic element:

2.7. Proposition. *Let E/K be an extension and $x \in E$. Assume that x is algebraic over K.*

1) *If $g \in K[X]$ is non-zero and such that $g(x) = 0$, then $f(x, K)$ divides g and $\deg f(x, K) \leq \deg g$.*
2) *$f(x, K)$ is irreducible in $K[X]$.*
3) *If F/K is a subextension of E/K, then $f(x, F)$ divides $f(x, K)$.*

Proof. Write $f = f(x, K)$ to denote the minimal polynomial of x. To prove 1) note that $g(x) = 0$ implies that g belongs to the kernel of the evaluation map, so g is a multiple of f. To prove 2) note that if $f = pq$ for some $p, q \in K[X]$ such that $0 < \deg p, \deg q < \deg f$, then $f(x) = 0$ implies that either $p(x) = 0$ or $q(x) = 0$, a contradiction. Finally, we prove 3). Since $f \in K[X] \subseteq F[X]$ and $f(x) = 0$, it follows from 1) that $f(x, F)$ divides f. $\square$

Some easy examples: $f(i, \mathbb{Q}) = X^2 + 1$, $f(i, \mathbb{C}) = X - i$ and $f(\sqrt[3]{2}, \mathbb{Q}) = X^3 - 2$. Here is the code:

```
> R<x> := PolynomialRing(Rationals());
> Q<i> := QuadraticField(-1);
> MinimalPolynomial(i);
x^2 + 1
```

A bit harder: $f(\sqrt[3]{2} + 1, \mathbb{Q}) = X^3 - 3X^2 + 3X - 3$. Here is the code:

```
> R<x> := PolynomialRing(Rationals());
> E<z> := NumberField(x^3-2);
> MinimalPolynomial(z+1);
x^3 - 3*x^2 + 3*x - 3
```

2.8. Example. Let us compute $f(\sqrt{2}+\sqrt{3},\mathbb{Q})$. Let $\alpha=\sqrt{2}+\sqrt{3}$. Then

$$\begin{aligned}\alpha-\sqrt{2}=\sqrt{3} &\implies (\alpha-\sqrt{2})^2=3 \implies \alpha^2-2\sqrt{2}\alpha+2=3\\ &\implies \alpha^2-1=2\sqrt{2}\alpha \implies (\alpha^2-1)^2=8\alpha^2 \implies \alpha^4-10\alpha^2+1=0.\end{aligned}$$

Thus α is a root of $g=X^4-10X^2+1$. To prove that $g=f(\alpha,\mathbb{Q})$ it is enough to prove that g is irreducible in $\mathbb{Q}[X]$. First note that the roots of g are $\sqrt{2}+\sqrt{3}$, $\sqrt{2}-\sqrt{3}$, $-\sqrt{2}+\sqrt{3}$ and $-\sqrt{2}-\sqrt{3}$. This means that if g is not irreducible, then $g=hh_1$ for some polynomials $h,h_1\in\mathbb{Q}[X]$ such that $\deg h=\deg h_1=2$. This is not possible, as

$$\begin{aligned}(\sqrt{2}+\sqrt{3})+(\sqrt{2}-\sqrt{3})&=2\sqrt{2}\notin\mathbb{Q},\\ (\sqrt{2}+\sqrt{3})+(-\sqrt{2}+\sqrt{3})&=2\sqrt{3}\notin\mathbb{Q},\\ (\sqrt{2}+\sqrt{3})(-\sqrt{2}-\sqrt{3})&=-5-2\sqrt{6}\notin\mathbb{Q}.\end{aligned}$$

2.9. Proposition. *Let F/K be a subextension of E/K. Then*

$$[E:K]=[E:F][F:K].$$

Proof. Let $\{e_i:i\in I\}$ be a basis of E over F and $\{f_j:j\in J\}$ be a basis of F over K. If $x\in E$, then $x=\sum_i\lambda_ie_i$ (finite sum) for some $\lambda_i\in F$. For each i, $\lambda_i=\sum_j a_{ij}f_j$ (finite sum) for some $a_{ij}\in K$. Then $x=\sum_i\sum_j a_{ij}(f_je_i)$. This means that $\{f_je_i:i\in I,j\in J\}$ generates E as a K-vector space. Let us prove that $\{f_je_i:i\in I,j\in J\}$ is linearly independent. If $\sum_i\sum_j a_{ij}(f_je_i)=0$ (finite sum) for some $a_{ij}\in K$, then

$$\begin{aligned}0=\sum_i\left(\sum_j a_{ij}f_j\right)e_i &\implies \sum_j a_{ij}f_j=0 \text{ for all } i\in I\\ &\implies a_{ij}=0 \text{ for all } i\in I \text{ and } j\in J.\end{aligned}$$ □

We state a lemma:

2.10. Lemma. *If A is a finite-dimensional commutative algebra over K and A is an integral domain, then A is a field.*

Proof. Let $a\in A\setminus\{0\}$. We need to prove that there exists $b\in A$ such that $ab=1$. Let $\theta\colon A\to A$, $x\mapsto ax$. Note that θ is a K-linear transformation, as

$$\theta(x+y)=a(x+y)=ax+ay=\theta(x)+\theta(y),\quad \theta(\lambda x)=a(\lambda x)=\lambda(ax)=\lambda\theta(x),$$

for all $x,y\in A$ and $\lambda\in K$. It is injective since A is an integral domain. Since $\dim_K A<\infty$, it follows that θ is an isomorphism. In particular, $\theta(A)=A$, which implies that there exists $b\in A$ such that $1=ab$. □

Let E/K be an extension and $x\in E$. Then

$$K[x]=\{f(x):f\in K[X]\}$$

is a subring of E that contains K. Note that $K[x]$ is a K-vector space.

More generally, if $x_1,\dots,x_n\in E$, then

$$K[x_1,\dots,x_n]=\{f(x_1,\dots,x_n):f\in K[X_1,\dots,X_n]\}$$

is a subring of E. Note that $K[x_1,\dots,x_n]$ is a K-vector space. Clearly, $K[x_1,\dots,x_n]$ is a domain and

$$K(x_1,\dots,x_n)=\left\{\frac{f(x_1,\dots,x_n)}{g(x_1,\dots,x_n)}:f,g\in K[X_1,\dots,X_n]\text{ with }g(x_1,\dots,x_n)\neq 0\right\}$$

is the extension of K generated by $x_1,\dots,x_n$. Note that

$$K(x_1,\dots,x_n)=(K(x_1,\dots,x_{n-1}))(x_n).$$

The previous construction can be generalized. Let I be a non-empty set. For each $i\in I$, let X_i be a variable. Consider the polynomial ring $K[\{X_i:i\in I\}]$ and let $S=\{x_i:i\in I\}$ be a subset of E. There exists a unique algebra homomorphism

$$K[\{X_i:i\in I\}]\to E$$

such that $X_i\mapsto x_i$ for all $i\in I$. The image is denoted by $K[S]$. In particular, an element $z\in K[S]$ is of the form

$$z=h(x_1,\dots,x_n)$$

for a polynomial $h\in K[X_1,\dots,X_n]$ in finitely many variables $X_1,\dots,X_n$ and $x_1,\dots,x_n\in S$.

2.11. Exercise. Prove that $\mathbb{Q}[\sqrt{2}]=\mathbb{Q}(\sqrt{2})$.

The exercise is not an accident.

2.12. Theorem. *Let E/K be an extension and $x\in E\setminus K$. The following statements are equivalent:*

- ***1)*** *x is algebraic over K.*
- ***2)*** $\dim_K K[x]<\infty$*.*
- ***3)*** *$K[x]$ is a field.*
- ***4)*** *$K[x]=K(x)$.*

Proof. We first prove 1) $\implies$ 2). Let $z\in K[x]$, say $z=h(x)$ for some $h\in K[X]$. There exists $g\in K[X]$ such that $g\neq 0$ and $g(x)=0$. Divide h by g to obtain polynomials $q,r\in K[X]$ such that $h=gq+r$, where $r=0$ or $\deg r<\deg g$. This implies that

$$z=h(x)=g(x)q(x)+r(x)=r(x).$$

If $\deg g=m$, then $r=\sum_{i=0}^{m-1}a_iX^i$ for some $a_0,\dots,a_{m-1}\in K$. Thus

$$z=\sum_{i=0}^{m-1}a_ix^i$$

and hence $K[x]\subseteq\langle 1,x,\dots,x^{m-1}\rangle$.

The previous lemma proves that 2) $\implies$ 3).

It is trivial that 3) $\implies$ 4).

It remains to prove that 4) $\implies$ 1). Since $x\neq 0$, $1/x\in K(x)=K[x]$. There exists $a_0,\dots,a_n\in K$ such that $1/x=a_0+a_1x+\cdots+a_nx^n$. Thus

$$a_nx^{n+1}+\cdots+a_1x^2+a_0x-1=0,$$

and hence x is a root of $a_nX^{n+1}+\cdots+a_0X-1\in K[X]\setminus\{0\}$. $\square$

Note that if x is algebraic over K, then $K[x] \simeq K[X]/(f(x,K))$.

2.13. Exercise. Let E/K be an extension and $x \in E$ be an algebraic element over K. Prove that the degree of x over K is equal to $[K(x):K]$.

2.14. Corollary. *If E/K is finite, then E/K is algebraic.*

Proof. Let $n = [E:K]$ and $x \in E \setminus K$. The set $\{1, x, \dots, x^n\}$ has $n+1$ elements, so it is linearly dependent. There exist $a_0, \dots, a_n \in K$, not all zero, such that

$$a_0 + a_1 x + \cdots + a_n x^n = 0.$$

Thus x is a root of the non-zero polynomial $a_0 + a_1 X + \cdots + a_n X^n \in K[X]$. $\square$

In Example 2.5 we proved that $\sqrt{2} + \sqrt[3]{3}$ and $\sqrt{2}\sqrt[3]{3}$ are algebraic over $\mathbb{Q}$. This can be easily proved now with Corollary 2.14.

2.15. Exercise. Let E/K be an extension and a and b be algebraic over K. Prove that $a+b$ and ab are algebraic over K.

We note that the converse of Corollary 2.14 does not hold.

2.16. Corollary. *If E/K is an extension and $x_1, \dots, x_n \in E$ are algebraic over K, then $K(x_1, \dots, x_n)/K$ is finite and $K(x_1, \dots, x_n) = K[x_1, \dots, x_n]$.*

Proof. We proceed by induction on n. The case $n = 1$ follows immediately from the theorem. So assume the result holds for some $n \geq 1$. Since the extensions

$$K(x_1, \dots, x_n)/K(x_1, \dots, x_{n-1}) \quad \text{and} \quad K(x_1, \dots, x_{n-1})/K$$

are both finite, it follows that $K(x_1, \dots, x_n)/K$ is finite. Moreover,

$$\begin{aligned} K(x_1, \dots, x_n) &= K(x_1, \dots, x_{n-1})(x_n) \\ &= K(x_1, \dots, x_{n-1})[x_n] = K[x_1, \dots, x_{n-1}][x_n] = K[x_1, \dots, x_n]. \end{aligned}$$ $\square$

2.17. Corollary. *Let $E = K(S)$ for some set S. Then E/K is algebraic if and only if x is algebraic over K for all $x \in S$.*

Proof. Let us prove the non-trivial implication. Let $z \in K(S)$. In particular, there exists a finite subset $T \subseteq S$ such that $z \in K(T)$. The previous result implies that $K(T)/K$ is algebraic, and hence z is algebraic. $\square$

If E/K is an extension, let

$$\overline{K}_E = \{x \in E : x \text{ is algebraic over } K\}.$$

2.18. Corollary. *If E/K is an extension, then $\overline{K}_E$ is a subfield of E that contains K. Moreover, $K(\overline{K}_E) = \overline{K}_E$ and $K(\overline{K}_E)/K$ is algebraic.*

Proof. By definition, $K(\overline{K}_E)/K$ is algebraic. Thus $K(\overline{K}_E) \subseteq \overline{K}_E$. From this, it follows that $K(\overline{K}_E) = \overline{K}_E$. $\square$

The following exercise is now almost trivial:

2.19. Exercise. Let E/K be an extension of finite type; this means that $E = K(S)$ for some finite set S. Prove that E/K is algebraic if and only if E/K is finite.

§ 2. Algebraic extensions

Let $\overline{\mathbb{Q}} = \{\alpha \in \mathbb{C} : \alpha \text{ is algebraic over } \mathbb{Q}\}$. Then $\overline{\mathbb{Q}}$ is the field of algebraic numbers. Can you compute $[\overline{\mathbb{Q}} : \mathbb{Q}]$?

2.20. Exercise. Prove that $[\mathbb{Q}[\sqrt[3]{2}] : \mathbb{Q}] = 3$.

For the previous exercise, you may use Eisenstein's criterion.

2.21. Exercise. Let $E = \mathbb{Q}[i, \sqrt{2}] = \mathbb{Q}[\sqrt{2}][i]$. Prove that $[E : \mathbb{Q}] = 4$.

2.22. Exercise. Let $E = \mathbb{Q}[\sqrt{2}, \sqrt[3]{5}]$.

1) Compute $[E : \mathbb{Q}]$.
2) Prove that $E = \mathbb{Q}[\sqrt{2} + \sqrt[3]{5}]$.
3) Find the minimal polynomial of $\sqrt{2} + \sqrt[3]{5}$ over $\mathbb{Q}$.

2.23. Exercise. Find the minimal polynomials of $\sqrt[4]{3}i$ over $\mathbb{Q}[i]$ and over $\mathbb{Q}[\sqrt{3}]$.

2.24. Exercise. Find the minimal polynomial of $\sqrt{2} + \sqrt[3]{5}i$ over $\mathbb{Q}[i]$.

Algebraic field extensions form a nice class of extensions. The same happens with finite field extensions.

2.25. Proposition. *Let F/K be a subextension of E/K. Then E/K is algebraic if and only if E/F and F/K are algebraic.*

Proof. If E/K is algebraic, then E/F and F/K are both algebraic, as $K \subseteq F \subseteq E$. Let us assume that E/F and F/K are both algebraic. Let $x \in E$ and let L be the subextension over K generated by the coefficients of $f(x, F)$, the minimal polynomial of x over F. Then L/K is finite, since it is generated by finitely many algebraic elements. Moreover, x is algebraic over L. Since

$$[L(x) : K] = [L(x) : L][L : K] < \infty,$$

$L(x)/K$ is algebraic. In particular, x is algebraic over K. □

2.26. Exercise. Let F/K be a subextension of E/K. Prove that E/K is finite if and only if E/F and F/K are finite.

Let K be a field and $K \subseteq F \subseteq L$ and $K \subseteq E \subseteq L$ be fields. The **composite** of E and F is defined as

$$EF = K(E \cup F) = F(E) = E(F)$$

and it is equal to the smallest field that contains E and F. Here is the picture:

$$
\begin{tikzcd}[arrows=dash]
& L \arrow[d] & \\
& EF \arrow[dl] \arrow[dr] & \\
E \arrow[dr] & & F \arrow[dl] \\
& K &
\end{tikzcd}
$$

2.27. Exercise. Let E/K and F/K be algebraic field extensions. Prove that

$$EF = \left\{ \sum_{i=1}^{m} e_i f_i : m \in \mathbb{Z}_{>0}, e_i \in E, f_i \in F \text{ for all } i \in \{1, \dots, m\} \right\}.$$

2.28. Exercise. If $E = \mathbb{Q}(\sqrt{2})$ and $F = \mathbb{Q}(\sqrt{3})$, then $EF = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Compute $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$ and $\mathbb{Q}(\sqrt{2}) \cap \mathbb{Q}(\sqrt{3})$.

2.29. Exercise. Let $\xi \in \mathbb{C}$ be a primitive cubic root of one. If $E = \mathbb{Q}(\sqrt[3]{2})$ and $F = \mathbb{Q}(\xi)$, then $EF = \mathbb{Q}(\sqrt[3]{2}, \xi)$. Compute $[\mathbb{Q}(\sqrt[3]{2}, \xi) : \mathbb{Q}]$ and $\mathbb{Q}(\sqrt[3]{2}) \cap \mathbb{Q}(\xi)$.

2.30. Exercise. Let E/K and F/K be extensions, where both E and F are subfields of a field L. If F/K is algebraic, then EF/E is algebraic.

2.31. Exercise. Let E/K and F/K be extensions, where both E and F are subfields of a field L. If F/K is finite, then EF/E is finite.

The solution to the previous exercise shows, in particular, that $[EF : E] \leq [F : K]$.

2.32. Lemma. *Let $\sigma\colon K \to L$ be a field homomorphism. Then there exists an extension E/K and a field isomorphism $\varphi\colon E \to L$ such that $\varphi|_K = \sigma$.*

Proof. Note that $\sigma\colon K \to \sigma(K)$ is bijective. Let A be a set in bijection with $L \setminus \sigma(K)$ and disjoint with K. Let $E = K \cup A$. If $\theta\colon A \to L \setminus \sigma(K)$ is bijective, then let

$$\varphi\colon E \to L, \quad \varphi(x) = \begin{cases} \sigma(x) & \text{if } x \in K, \\ \theta(x) & \text{if } x \in A. \end{cases}$$

Then φ is a bijective map such that $\varphi|_K = \sigma$. Transport the operations of L onto E, that is to define binary operations on E as follows:

$$(x, y) \mapsto x \oplus y = \varphi^{-1}(\varphi(x) + \varphi(y)), \qquad (x, y) \mapsto x \odot y = \varphi^{-1}(\varphi(x)\varphi(y)).$$

Then, for example,

$$x \oplus y = \varphi^{-1}(\varphi(x) + \varphi(y)) = \varphi^{-1}(\sigma(x) + \sigma(y)) = \varphi^{-1}(\sigma(x + y)) = \varphi^{-1}(\varphi(x + y)) = x + y$$

for all $x, y \in K$. $\square$

If $\sigma\colon A \to B$ is a ring homomorphism, then σ induces a ring homomorphism

$$\overline{\sigma}\colon A[X] \to B[X], \quad \sum_i a_i X^i \mapsto \sum_i \sigma(a_i) X^i.$$

2.33. Theorem. *Let K be a field and $f \in K[X]$ be such that $\deg f > 0$. Then there exists an extension E/K such that f admits a root in E.*

Proof. We may assume that f is irreducible over K. Let $L = K[X]/(f)$ and

$$\pi\colon K[X] \to L$$

be the canonical map. Then L is a field (the reader should explain why). Let $\sigma\colon K \to L$, $a \mapsto \pi(aX^0)$, and $g = \overline{\sigma}(f) \in L[X]$.

We claim that $\pi(X)$ is a root of g in L. Suppose that $f = \sum_i a_i X^i$. Then

$$\begin{aligned} g(\pi(X)) &= \overline{\sigma}(f)(\pi(X)) \\ &= \sum_i \sigma(a_i)\pi(X)^i = \sum_i \pi(a_i X^0)\pi(X^i) = \pi(\sum a_i X^i) = \pi(f) = 0. \end{aligned}$$

By Lemma 2.32, there exists an extension E/K and an isomorphism $\varphi\colon E \to L$ such that $\varphi|_K = \sigma$. Note that $\varphi(x) = 0$ if and only if $x = 0$. If $u = \pi(X)$, then $\varphi^{-1}(u)$ is a root of f in E, as

$$\begin{aligned} \varphi(f(\varphi^{-1}(u))) &= \varphi\left(\sum_i a_i \varphi^{-1}(u)^i\right) = \varphi\left(\sum_i a_i \varphi^{-1}(u^i)\right) \\ &= \sum_i \varphi(a_i) u^i = \sum_i \sigma(a_i) u^i = g(u) = 0. \end{aligned}$$ □

As a corollary, if K is a field and $f_1, \dots, f_n \in K[X]$ are polynomials of positive degree, then there exists an extension E/K such that each f_i admits a root in E. This is proved by induction on n.

2.34. Definition. A field K is **algebraically closed** if each $f \in K[X]$ of positive degree admits a root in K.

The **fundamental theorem of algebra** states that $\mathbb{C}$ is algebraically closed. A typical proof uses complex analysis. Later we will give a proof of this result using Galois theory.

2.35. Proposition. *The following statements are equivalent:*

1) *K is algebraically closed.*
2) *If $f \in K[X]$ is irreducible, then $\deg f = 1$.*
3) *If $f \in K[X]$ is non-zero, then f decomposes linearly in $K[X]$, that is*

$$f = a \prod_{i=1}^{n} (X - \alpha_i)^{m_i}$$

for some $a \in K$ and $\alpha_1, \dots, \alpha_n \in K$.
4) *If E/K is algebraic, then $E = K$.*

Proof. 1) $\implies$ 2) $\implies$ 3) are exercises.

Let us prove that 3) $\implies$ 4). Let $x \in E$. Decompose $f(x,K)$ linearly in $K[X]$ as

$$f(x,K) = a\prod_{i=1}^{n}(X-\alpha_i)^{m_i}$$

and evaluate on x to obtain that $x = \alpha_j$ for some j.

To prove that 4) $\implies$ 1) let $f \in K[X]$ be such that $\deg f > 0$. There exists an extension E/K such that f has a root x in E. The extension $K(x)/K$ is algebraic and hence $K(x) = K$, so $x \in K$. □

§ 3. Artin's theorem

3.1. Definition. An **algebraic closure** of a field K is an algebraic extension C/K such that C is algebraically closed.

For example, $\mathbb{C}/\mathbb{R}$ is an algebraic closure but $\mathbb{C}/\mathbb{Q}$ is not.

3.2. Proposition. *Let C be algebraically closed and $\sigma\colon K \to C$ be a field homomorphism. If E/K is algebraic, then there exists a field homomorphism $\varphi\colon E \to C$ such that $\varphi|_K = \sigma$.*

Proof. Suppose first that $E = K(x)$ and let $f = f(x,K)$. Let $\overline{\sigma}(f) \in C[X]$ and let $y \in C$ be a root of $\overline{\sigma}(f)$. If $z \in E$, then $z = g(x)$ for some $g \in K[X]$. Let $\varphi\colon E \to C$, $z \mapsto \overline{\sigma}(g)(y)$.

The map φ is well-defined. If $z = h(x)$ for some $h \in K[X]$, then

$$0 = g(x) - h(x) = (g-h)(x)$$

and thus f divides $g-h$. In particular, $\overline{\sigma}(f)$ divides $\overline{\sigma}(g-h) = \overline{\sigma}(g) - \overline{\sigma}(h)$ and hence

$$(\overline{\sigma}(g) - \overline{\sigma}(h))(y) = 0.$$

It is an exercise to show that the map φ is a ring homomorphism.

Let $a \in K$. It follows that $\varphi|_K = \sigma$, as

$$\varphi(a) = \overline{\sigma}(aX^0)(y) = \sigma(a)$$

Let us now prove the proposition in full generality. Let X be the set of pairs (F,τ), where F is a subfield of E that contains K and $\tau\colon F \to C$ is a field homomorphism such that $\tau|_K = \sigma$. Note that $(K,\sigma) \in X$, so X is non-empty. Moreover, X is partially ordered by

$$(F,\tau) \leq (F_1,\tau_1) \iff F \subseteq F_1 \text{ and } \tau_1|_F = \tau.$$

If $\{(F_i,\tau_i) : i \in I\}$ is a chain in X, then $F = \cup_{i\in I}F_i$ is a subfield of E that contains K. Moreover, if $z \in F$, then $z \in F_i$ for some $i \in I$ and then one defines $\tau(z) = \tau_i(z)$. It is an exercise to prove that τ is well-defined. Since $(F,\tau) \in X$ is an upper bound, Zorn's lemma implies that there exists a maximal element $(E_1,\theta) \in X$. We claim that $E = E_1$. If not, let $z \in E \setminus E_1$. Since we know the proposition is true for the extension $E_1(z)/E_1$, let $\rho\colon E_1(z) \to C$ be a field homomorphism such that $\rho|_{E_1} = \theta$. Then, in particular, $\rho|_K = \sigma$. This implies that $(E_1(z),\rho) \in X$ and hence $(E_1,\theta) < (E_1(z),\rho)$, a contradiction to the maximality of (E_1,θ). □

The previous proposition will be used to prove that the algebraic closure is unique up to isomorphism.

3.3. Theorem (Artin). *Let K be a field. Then K admits an algebraic closure C/K. If C_1/K is an algebraic closure, then the extensions C/K and C_1/K are isomorphic.*

PROOF. Let us first prove the uniqueness. The previous proposition implies the existence of an extension homomorphism $\varphi\colon C \to C_1$. Let $y \in C_1$ and $f = f(y, K)$ be the minimal polynomial of y in K. Since f admits a factorization

$$f = \prod (X - \alpha_i)^{m_i}$$

in $C[X]$, it follows that

$$f = \overline{\varphi}(f) = \prod (X - \varphi(\alpha_i))^{m_i}$$

Since $0 = f(y)$, we conclude that $y = \varphi(\alpha_j)$ for some j. In particular, φ is surjective and hence φ is bijective.

We now prove the existence. Let us assume that K admits an extension E/K with E algebraically closed. We will prove later that this extension indeed exists; at the moment, we only want to get an algebraic extension from this setting. Let

$$F = \{x \in E : x \text{ is algebraic over } K\}.$$

Then F/K is algebraic. Let $g \in F[X]$ be such that $\deg g > 0$. Since E is algebraically closed, g admits a root α in E. In particular, α is algebraic over F and hence α is algebraic over K. This implies that $\alpha \in F$, thus F is algebraically closed. This proves that F/K is an algebraic closure.

Let us prove that there exists an extension E_1/K such that every polynomial $f \in K[X]$ with $\deg f > 0$ has a root in E_1. Let $\{f_i : i \in I\}$ be the family of monic irreducible polynomials with coefficients in K. We may think that $f_i = f_i(X_i)$. Let $R = K[\{X_i : i \in I\}]$ and let J be the ideal of R generated by the $f_i(X_i)$. We claim that $J \neq R$. If not, $1 \in J$, so

$$1 = \sum_{j=1}^{m} g_j f_{i_j}(X_{i_j})$$

for some $g_1, \dots, g_m \in R$. There exists an extension F/K such that f_{i_j} has a root α_j in F for all j. Let

$$\tau\colon R \to F, \quad \tau(X_k) = \begin{cases} \alpha_j & \text{if } k = i_j, \\ 0 & \text{if } k \notin \{i_1, \dots, i_m\}. \end{cases}$$

Then τ is a ring homomorphism and

$$1 = \tau(1) = \sum_{j=1}^{m} \tau(g_j) f_{i_j}(\alpha_j) = 0,$$

a contradiction.

Since J is a proper ideal, it is contained in a maximal ideal M. Let $L = R/M$ and let $\sigma\colon K \to L$ be the composition $K \hookrightarrow R \to R/M = L$, where $\pi\colon R \to R/M$ is the canonical map. As we did before, $\pi(X_i)$ is a root of $\overline{\sigma}(f_i)$ for all i. And there exists an extension E_1/K such that every f_i has a root in E_1. Proceeding in this way, we construct a sequence

$$E_1 \subseteq E_2 \subseteq \cdots$$

of fields such that every polynomial of positive degree and coefficients in E_k admits a root in E_{k+1}. Let $E = \cup E_k$. We claim that E is algebraically closed. In fact, let $g \in E[X]$ be such that $\deg g > 0$. Then, since $g \in E_r[X]$ for some r, it follows that g has a root in $E_{r+1} \subseteq E$. $\square$

§ 4. Splitting fields

4.1. Definition. Let K be a field and $f \in K[X]$ be such that $\deg f > 0$. A **splitting field** of f over K is a field E that contains K and that satisfies the following properties:

1) f factorizes linearly in $E[X]$.
2) If F is a field such that $K \subseteq F \subseteq E$ and f factorizes linearly in $F[X]$, then $F = E$.

Easy examples:

4.2. Example. $\mathbb{C}$ is a splitting field of $X^2 + 1 \in \mathbb{R}[X]$.

4.3. Example. $\mathbb{Q}[\sqrt{2}]$ is a splitting field of $X^2 - 2 \in \mathbb{Q}[X]$.

4.4. Example. The splitting field of $f = X^2 - 2$ over $\mathbb{Z}/7$ is precisely $\mathbb{Z}/7$, as 3 and 4 are the roots of f in $\mathbb{Z}/7$.

4.5. Example. $\mathbb{Q}(\sqrt[3]{2})$ is not a splitting field of $X^3 - 2 \in \mathbb{Q}[X]$. However, if ω is a primitive cubic root of one, then $\mathbb{Q}(\sqrt[3]{2}, \omega)$ is a splitting field of the polynomial $X^3 - 2 \in \mathbb{Q}[X]$.

4.6. Proposition. *E is a splitting field of $f \in K[X]$ if and only if f factorizes linearly in $E[X]$ and $E = K(x_1, \dots, x_n)$, where $x_1, \dots, x_n$ are the roots of f.*

Proof. Let $f = a \prod_{i=1}^{r} (X - x_i)^{n_i}$ and $F = K(x_1, \dots, x_r)$ with $x_1, \dots, x_r \in E$. Since f factorizes linearly in $F[X]$, it follows that $F = E$. Conversely, let $E = K(x_1, \dots, x_r)$ and assume that f factorizes linearly in $F[X]$. Then, in particular, $x_1, \dots, x_r \in F$. Hence $E \subseteq F$ and $F = E$. $\square$

Note that if $E = K(x_1, \dots, x_n)$ as in the statement of Proposition 4.6, then it automatically follows that f factorizes linearly in $E[X]$.

One immediately obtains the following consequence: If E is a splitting field of $f \in K[X]$, then E/K is finite.

4.7. Theorem. *Let $f \in K[X]$ be such that $\deg f > 0$. There exists a (unique up to extension isomorphism) splitting field of f over K.*

Proof. Let C/K be an algebraic closure of K. Write

$$f = a \prod_{i=1}^{r} (X - x_i)^{n_i}$$

in $C[X]$. Then $E = K(x_1, \dots, x_r)$ is a splitting field of f over K.

Let us prove the uniqueness: if E_1/K is a splitting field of f over K, then E_1/K is algebraic and thus Proposition 3.2 implies that there exists $\varphi \in \operatorname{Hom}(E_1/K, C/K)$, that is $\varphi\colon E_1 \to C$ is a field homomorphism such that $\varphi|_K$ is the identity. Factorize f linearly in $E_1[X]$ and apply $\overline{\varphi}$:

$$f = a \prod_{j=1}^{s} (X - y_j)^{m_j} \implies f = \overline{\varphi}(f) = \varphi(a) \prod_{j=1}^{s} (X - \varphi(y_j))^{m_j}$$

so f factorizes linearly in $\varphi(E_1)[X]$. Moreover, $E_1 = K(y_1, \dots, y_s)$ and

$$\varphi(E_1) = K(\varphi(y_1), \dots, \varphi(y_s)).$$

Thus $\varphi(E_1)$ is a splitting field of f. Since $\varphi(E_1) \subseteq E$, it follows that $\varphi(E_1) = E$. $\square$

4.8. Exercise. If C is an algebraic closure of K and $\varphi \in \mathrm{Hom}(C/K, C/K)$, then φ is an isomorphism.

Let C be an algebraic closure of K and $G = \mathrm{Gal}(C/K)$. The group G acts on C

$$\sigma \cdot x = \sigma(x), \quad \sigma \in G,\, x \in C.$$

The orbits are of the form

$$O_G(x) = \{\sigma(x) : \sigma \in G\} = \{y \in C : y = \sigma(x) \text{ for some } \sigma \in G\}$$

The elements $x, y \in C$ are **conjugate** if $y = \sigma(x)$ for some $\sigma \in G$.

4.9. Proposition. *Let C be an algebraic closure of K and $x, y \in C$. Then x and y are conjugate if and only if $f(x,K) = f(y,K)$. In particular, $O_G(x)$ is finite.*

Proof. Let $G = \mathrm{Gal}(C/K)$. If x and y are conjugate, say $y = \sigma(x)$ for some $\sigma \in G$, let us write $g = f(x,K)$ as

$$g = \sum_{i=0}^{n} a_i X^i$$

for some $n \geq 1$ and $a_0, \dots, a_n \in K$. Then $0 = g(x) = \sum_{i=0}^{n} a_i x^i$ and hence y is a root of g, as

$$\begin{aligned} 0 = \sigma\left(\sum_{i=0}^{n} a_i x^i\right) &= \sum_{i=0}^{n} \sigma(a_i)\sigma(x)^i \\ &= \sum_{i=0}^{n} a_i \sigma(x)^i = \sum_{i=0}^{n} a_i y^i. \end{aligned}$$

Thus $f(y,K) = g$.

Conversely, assume that $f(x,K) = f(y,K)$. Let $g = f(x,K) = f(y,K)$ and let

$$\varphi \colon K[x] \to K[y], \quad h(x) \mapsto h(y).$$

Let us show that the map φ is well-defined: we need to show that if $h_1(x) = h_2(x)$, then

$$h_1(y) = \varphi(h_1(x)) = \varphi(h_2(x)) = h_2(y).$$

If $h_1(x) = h_2(x)$, then

$$(h_1 - h_2)(x) = h_1(x) - h_2(x) = 0.$$

This implies that g divides $h_1 - h_2$. In particular, $h_1(y) = h_2(y)$.

A straightforward calculation shows that φ is a field homomorphism such that $\varphi|_K = \mathrm{id}$, this means that φ is an extension homomorphism such that $\varphi(x) = y$. There exists a homomorphism $\sigma \in \mathrm{Hom}(C/K, C/K)$ such that $\sigma|_{K[x]} = \varphi$. Since σ is bijective (why?), $\sigma(x) = \varphi(x) = y$ and hence $O_G(x) = O_G(y)$. $\square$

4.10. Proposition. *Let C be an algebraic closure of K and $x \in C$. Then*

$$f(x,K) = \prod_{y \in O_G(x)} (X - y)^m$$

for some m.

Proof. For each $y \in O_G(x)$ let m_y be the multiplicity of y in $f(x,K)$. Then, for example, $f(x,K) = (X-x)^{m_x} g$ for some g. If $y \in O_G(x)$, then $y = \sigma(x)$ for some $\sigma \in \mathrm{Gal}(C/K)$. Since

$$\overline{\sigma}(f(x,K)) = f(x,K) = (X-y)^{m_x}\overline{\sigma}(g),$$

it follows that $m_y \geq m_x$. By symmetry, we conclude that $m_x = m_y$. □

The previous proposition shows, in particular, that all the roots of an irreducible polynomial $f \in K[X]$ in an algebraic closure C of K have the same multiplicity. This is not true if f is not irreducible. Find an example.

4.11. Definition. Let K be a field and $\{f_i : i \in I\}$ be a non-empty family of polynomials of positive degree with coefficients in K. A **splitting field** of $\{f_i : i \in I\}$ is an extension E/K such that every f_i factorizes linearly in $E[X]$ and if F/K is a subextension of E/K such that every f_i factorizes linearly in $F[X]$, then $F = E$.

4.12. Exercise. Prove that E/K is a splitting field of $\{f_i : i \in I\}$ if and only if every f_i factorizes linearly in $E[X]$ and $E = K(S)$ where $S = \{\text{roots of } f_i \text{ for all } i\}$.

4.13. Exercise. Prove that if E/K is a splitting field of $\{f_i : i \in I\}$, then E/K is algebraic. If, moreover, I is finite, then E/K is a splitting field of $\prod_{i\in I} f_i$.

4.14. Exercise. Prove that there exists a splitting field of $\{f_i : i \in I\}$ and it is unique up to extension isomorphism.

4.15. Exercise. Let $f = X^3 - X - 1 \in (\mathbb{Z}/3)[X]$ and E be a splitting field of f. Compute $[E : \mathbb{Z}/3]$.

What about the splitting field of $f = X^3 - X - 1 \in \mathbb{Q}[X]$?

4.16. Exercise. Let $f = X^4 - 5X^2 + 5 \in \mathbb{Q}[X]$ and E be a splitting field of f. Compute $[E : \mathbb{Q}]$ and $\mathrm{Gal}(E/\mathbb{Q})$.

§ 5. Normal extensions

5.1. Proposition. *Let E/K be an algebraic extension and $\sigma \in \mathrm{Hom}(E/K, E/K)$. Then σ is bijective.*

Proof. It is enough to prove that σ is surjective. Why?

Let $x \in E$, $f = f(x,K)$, and let $S = \{y \in E : f(y) = 0\}$. Note that for any $y \in S$, we have that $\sigma(y) \in S$, and hence $\sigma(S) \subseteq S$. However, as σ is injective and S is finite, we must have that $\sigma(S) = S$. Hence there exists a $y \in S \subseteq E$ such that $\sigma(y) = x$, and therefore σ must be surjective.

□

5.2. Definition. Let E/K be an algebraic extension and C be an algebraic closure of K containing E. Then E/K is **normal** if $\sigma(E) \subseteq E$ for all $\sigma \in \mathrm{Hom}(E/K, C/K)$.

Note that $\sigma(E) \subseteq E$ in the previous definition is equivalent to $\sigma(E) = E$.

5.3. Example. The extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not normal. Why?

Some trivial examples of normal extensions: K/K is normal and if C is an algebraic closure of K, then C/K is normal.

5.4. Example. The extension $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is normal. Every extension generated by algebraic elements of degree two is normal.

5.5. Exercise. Let ξ be a primitive cubic root of one. Then $\mathbb{Q}(\sqrt[3]{2},\xi)/\mathbb{Q}$ is normal.

The following result is practical but technical. That is why we leave the proof as an exercise.

5.6. Exercise. Prove that the previous definition depends only on E (and not on the algebraic closure C).

Some properties:

5.7. Proposition. *Let E/K be a normal extension and $f\in K[X]$ be an irreducible polynomial that admits a root x in E. Then f factorizes linearly in E.*

Proof. We may assume that f is monic. Let C/K be an algebraic closure of K containing E. Let y be a root of f in C. Since $f=f(x,K)=f(y,K)$, it follows that $y=\sigma(x)$ for some $\sigma\in\mathrm{Gal}(C/K)$. Since E/K is normal, $\sigma|_E\colon E\to C$ is an automorphism of E/K, that is $\sigma(E)\subseteq E$. In particular, $y\in E$. □

Let $K\subseteq F\subseteq E$ be a tower of fields. If E/K is normal, then E/F is normal. However, Note that E/K normal does not imply F/K normal, as this would imply that every extension is normal. Moreover, E/F normal and F/K normal do not imply E/K normal.

5.8. Example. The extensions $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ are both normal, but the extension $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is not, as the roots of X^4-2 are $\sqrt[4]{2}$, $-\sqrt[4]{2}$, $\sqrt[4]{2}i$ and $-\sqrt[4]{2}i$.

Recall that if C is an algebraic closure of K and $x\in C$, then

$$f(x,K)=\prod(X-y)^m,$$

where the product is taken over all $y\in O_{\mathrm{Gal}(C/K)}(x)$. If E/K is normal and $x\in E$, then there exists m such that

$$f(x,K)=\prod(X-y)^m,$$

where the product is taken over all $y\in O_{\mathrm{Gal}(E/K)}(x)$.

5.9. Proposition. *Let E/K and F/K be extensions. If F/K is normal, then EF/E is normal.*

Proof. Let C be an algebraic closure of E containing EF (this exists because EF/E is algebraic). Let $\sigma\in\mathrm{Hom}(EF/E,C/E)$. We claim that $\sigma(EF)=EF$. Let

$$\overline{K}=\{x\in C : x\text{ is algebraic over }K\}.$$

Then $\overline{K}$ is an algebraic closure over K and $F\subseteq\overline{K}$. Since F/K is normal and $\sigma|_F\in\mathrm{Hom}(F/K,\overline{K}/K)$, it follows that $\sigma(F)=F$. If $z\in EF$, then $z=\sum_{i=1}^m e_if_i$ for some

$e_1, \dots, e_m \in E$ and $f_1, \dots, f_m \in F$. Since $\sigma(e_i) = e_i$ for all i,

$$\sigma(z) = \sum_{i=1}^{m} \sigma(e_i)\sigma(f_i) = \sum_{i=1}^{m} e_i\sigma(f_i) \in EF. \qquad \square$$

What is the relation between normal extensions and splitting fields? The notions look deeply related. The following proposition serves as an explanation:

5.10. Proposition. *Let E/K be an algebraic extension. Then E/K is normal if and only if E/K is the splitting field of a family of polynomials of $K[X]$ of positive degree.*

Proof. Assume first that E/K is a normal extension. Let $G = \mathrm{Gal}(E/K)$. If $x \in E$ and $f(x,K) = \prod_{y \in O_G(x)}(X-y)^m$, then $f(x,K)$ factorizes linearly in $E[X]$. Thus E/K is a splitting field of the family $\{f(x,K) : x \in E\}$.

Conversely, assume that E/K is a splitting field of the family $\{f_i : i \in I\}$. Then $E = K(S)$ where S is the set of roots of the polynomials f_i. Let C/K be an algebraic closure of K that contains E and let $\sigma \in \mathrm{Hom}(E/K, C/K)$. Let $x \in S$. Then x is a root of some $f_j = \sum a_k X^k$. Since $f_j(x) = 0$, it follows that $\sigma(x)$ is a root of f_j, as

$$f_j(\sigma(x)) = \sum a_k \sigma(x)^k = \sum \sigma(a_k)\sigma(x^k) = \sigma\left(\sum a_k x^k\right) = \sigma(0) = 0.$$

Hence $\sigma(E) \subseteq E$. $\square$

5.11. Exercise. Let $E = \mathbb{Q}[\sqrt[4]{7} + \sqrt{2}]$.

1) Prove that $E/\mathbb{Q}$ is not normal.
2) Compute $[E : \mathbb{Q}]$.
3) Compute $\mathrm{Gal}(E/\mathbb{Q})$.

§ 6. Dedekind's theorem

Note that every extension homomorphism $E/K \to F/K$ is, in particular, a K-linear map $E \to F$, that is

$$\mathrm{Hom}(E/K, F/K) \subseteq \mathrm{Hom}_K(E, F).$$

If F/K is an extension and V is a K-vector space, the set $\mathrm{Hom}_K(V, F)$ of K-linear maps is a vector space over F with $(a \cdot f)(v) = af(v)$ for $a \in F$, $f \in \mathrm{Hom}_K(V, F)$ and $v \in V$.

6.1. Exercise. Let V be a K-vector space. Prove that $\dim_F \mathrm{Hom}_K(V, F) \geq \dim_K V$. Moreover, if $\dim_K V < \infty$, then $\dim_F \mathrm{Hom}_K(V, F) = \dim_K V$.

If V is a vector space and S is a (possibly infinite) subset of V, then S is linearly independent if every finite subset of S is linearly independent.

6.2. Theorem (Dedekind). *Let E/K and F/K be extensions and let $\{\varphi_i : i \in I\}$ be a subset of* $\mathrm{Hom}(E/K, F/K)$*, i.e. a family of extension homomorphisms. Assume that $\varphi_i \neq \varphi_j$ if $i \neq j$. Then the subset $\{\varphi_i : i \in I\} \subseteq \mathrm{Hom}_K(E, F)$ is linearly independent over F.*

Proof. Assume it is not. Let $\{\varphi_1, \dots, \varphi_n\}$ be linearly dependent over F with n minimal. Clearly, $n > 1$. Without loss of generality, we may assume that

$$\sum_{i=1}^{n} a_i \varphi_i = 0 \tag{6.1}$$

for some $a_1,\dots,a_n\in F$ all different from zero. Let $z\in E\setminus\{0\}$ be such that $\varphi_1(z)\neq\varphi_2(z)$. If $x\in E$, then

$$0=\left(\sum_{i=1}^n a_i\varphi_i\right)(xz)=\sum_{i=1}^n a_i\varphi_i(xz)=\sum_{i=1}^n a_i\varphi_i(x)\varphi_i(z)=\left(\sum_{i=1}^n (a_i\varphi_i(z))\varphi_i\right)(x).$$

Thus

$$\sum_{i=1}^n (a_i\varphi_i(z))\varphi_i=0.$$

Since $\varphi_1(z)\neq 0$,

$$a_1\varphi_1+a_2\frac{\varphi_2(z)}{\varphi_1(z)}\varphi_2+\cdots+a_n\frac{\varphi_n(z)}{\varphi_1(z)}\varphi_n=0. \tag{6.2}$$

Thus, subtracting (6.1) and (6.2),

$$\left(a_2-a_2\frac{\varphi_2(z)}{\varphi_1(z)}\right)\varphi_2+\cdots+\left(a_n-a_n\frac{\varphi_n(z)}{\varphi_1(z)}\right)\varphi_n=0.$$

Since $a_2\neq 0$ and $\varphi_2(z)\neq\varphi_1(z)$, the scalar $a_2-a_2\frac{\varphi_2(z)}{\varphi_1(z)}\neq 0$ and hence $\{\varphi_2,\dots,\varphi_n\}$ is linearly dependent, a contradiction. □

If E/K and F/K are extensions, let $\gamma(E/K,F/K)=|\operatorname{Hom}(E/K,F/K)|$.

6.3. Exercise. Prove the following statements:

1) $\gamma(E/K,F/K)\leq\dim_F\operatorname{Hom}_K(E,F)$.
2) If $[E:K]<\infty$, then $\gamma(E/K,F/K)\leq[E:K]$.
3) If x is algebraic over K, then $\gamma(K(x)/K,F/K)\leq\deg f(x,K)$.

If C is an algebraic closure of K, then we define $\gamma(E/K)=\gamma(E/K,C/K)$. This definition does not depend on the algebraic closure.

6.4. Exercise. If C and C_1 are algebraic closures of K, then

$$|\operatorname{Hom}(E/K,C/K)|=|\operatorname{Hom}(E/K,C_1/K)|.$$

6.5. Proposition. *Let C be an algebraic closure of K and $G=\operatorname{Gal}(C/K)$. If $x\in C$, then $\gamma(K(x)/K)=|O_G(x)|$.*

Proof. If $\sigma\in\operatorname{Hom}(K(x)/K,C/K)$, then there exists $\phi\in G$ such that $\phi|_{K(x)}=\sigma$. Thus

$$\sigma(x)=\phi(x)\in O_G(x).$$

Conversely, if $y\in O_G(x)$, then there exists $\tau\in G$ such that $y=\tau(x)$. Hence

$$\tau|_{K(x)}\in\operatorname{Hom}(K(x)/K,C/K)$$

and $\tau|_{K(x)}(x)=y$. Since our sets are then in bijective correspondence, the claim follows. □

6.6. Exercise. If E/K is finite, then $|\operatorname{Gal}(E/K)|\leq[E:K]$. Moreover, E/K is normal if and only if $|\operatorname{Gal}(E/K)|=\gamma(E/K)$.

If $t\colon A \to B$ is a surjective map, then $a \sim a_1 \iff t(a) = t(a_1)$ defines an equivalence relation on A. The set $\overline{A}$ of equivalence classes is in bijective correspondence with B, $\overline{A} \to B$, $\overline{a} \mapsto t(a)$. Moreover, if $|t^{-1}(\{b\})| = m$ for all $b \in B$, then $|A| = m|\overline{A}| = m|B|$.

6.7. Proposition. *Let E/K be algebraic and F/K be a subextension such that E/F is finite. Then $\gamma(E/K) = \gamma(E/F)\gamma(F/K)$.*

Proof. Assume first that $E = F(x)$. Let C be an algebraic closure of K containing E and $G = \mathrm{Gal}(C/F)$. Let $f = f(x, F) = \sum b_i X^i$.

The map

$$\lambda\colon \mathrm{Hom}(E/K, C/K) \to \mathrm{Hom}(F/K, C/K), \quad \sigma \mapsto \sigma|_F,$$

is well-defined. It is surjective: if $\varphi \in \mathrm{Hom}(F/K, C/K)$, then $\varphi\colon F \to C$ is, in particular, a field homomorphism. Since E/F is algebraic, by Proposition 3.2 there exists a field homomorphism $\sigma\colon E \to C$ such that $\sigma|_F = \varphi$. Since $\sigma|_K = \varphi|_K = \mathrm{id}$, in particular $\sigma \in \mathrm{Hom}(E/K, C/K)$.

For $\varphi \in \mathrm{Hom}(F/K, C/K)$,

$$\lambda^{-1}(\{\varphi\}) = \{\sigma \in \mathrm{Hom}(E/K, C/K) : \sigma|_F = \varphi\}$$

and let R_φ be the set of roots (in C) of the polynomial $\overline{\varphi}(f) = \sum \varphi(b_i) X^i$.

Claim. The map $\alpha\colon \lambda^{-1}(\{\varphi\}) \to R_\varphi$, $\sigma \mapsto \sigma(x)$, is well-defined.

We need to show that $\sigma(x)$ is a root of $\overline{\varphi}(f)$:

$$\begin{aligned} \overline{\varphi}(f)(\sigma(x)) &= \sum \varphi(b_i)\sigma(x)^i = \sum \sigma(b_i)\sigma(x^i) \\ &= \sum \sigma(b_i x^i) = \sigma\left(\sum b_i x^i\right) = \sigma(f(x)) = \sigma(0) = 0. \end{aligned}$$

Claim. The map $\beta\colon R_\varphi \to \lambda^{-1}(\{\varphi\})$, $y \mapsto \sigma_y$, where $\sigma_y(z) = \overline{\varphi}(h)(y)$ if $z = h(x)$, is well-defined.

We need to show that if $z = h(x)$ and $z = h_1(x)$ for some $h, h_1 \in F[X]$, then

$$\overline{\varphi}(h)(y) = \overline{\varphi}(h_1)(y).$$

The assumptions imply that $(h - h_1)(x) = 0$ and hence f divides $h - h_1$. Since $\overline{\varphi}$ is a ring homomorphism, $\overline{\varphi}(f)$ divides $\overline{\varphi}(h) - \overline{\varphi}(h_1)$. This implies $(\overline{\varphi}(h) - \overline{\varphi}(h_1))(y) = 0$. We also need to show that $\sigma_y|_F = \varphi$: if $a \in F$, then write $a = aX^0 \in F[X]$. Thus $\sigma_y(a) = \overline{\varphi}(aX^0)(y) = \varphi(a) \in C$. It is now an exercise to prove that $\sigma_y \in \mathrm{Hom}(E/K, C/K)$.

Claim. $|\lambda^{-1}(\{\varphi\})| = |R_\varphi|$.

For this we need to show that β is the inverse of α, that is $\alpha \circ \beta = \mathrm{id}$ and $\beta \circ \alpha = \mathrm{id}$. To prove that $\beta \circ \alpha = \mathrm{id}$ let σ be such that $\sigma|_F = \varphi$. Then $y = \sigma(x) \in R_\varphi$. Let

$$z = h(x) = \sum a_i x^i \in F[x] = E.$$

Then

$$\overline{\varphi}(h)(y) = \sum \varphi(a_i) y^i = \sum \sigma(a_i) y^i = \sigma\left(\sum a_i x^i\right) = \sigma(z).$$

Conversely, if $y \in R_\varphi$, then

$$\alpha(\sigma_y) = \sigma_y(x) = y,$$

as $\sigma_y(x) = \overline{\varphi}(X)(y) = y$.

Claim. If $\phi \in \mathrm{Gal}(C/K)$ is such that $\phi|_F = \varphi$, then $|\phi^{-1}(R_\varphi)| = |R_\varphi|$ and

$$O_G(x) = \phi^{-1}(R_\varphi).$$

Let us first prove $O_G(x) \supseteq \phi^{-1}(R_\varphi)$. If $y \in R_\varphi$, then

$$\begin{aligned} f(\phi^{-1}(y)) &= \sum b_i \phi^{-1}(y^i) = \phi^{-1}\left(\sum \phi(b_i) y^i\right) \\ &= \phi^{-1}(\overline{\varphi}(f)(y)) = \phi^{-1}(0) = 0. \end{aligned}$$

Then $f(x, F) = f(\phi^{-1}(y), F)$. By Proposition 4.9, $\phi^{-1}(y) \in O_G(x)$.

Now we prove $O_G(x) \subseteq \phi^{-1}(R_\varphi)$. Let $z \in O_G(x)$. Then $\overline{\varphi}(f)(\phi(z)) = 0$, as

$$\begin{aligned} \overline{\varphi}(f)(\phi(z)) &= \sum \varphi(b_i)\phi(z^i) \\ &= \sum \phi(b_i)\phi(z^i) = \phi\left(\sum b_i z^i\right) = \phi(f(z)) = \phi(0) = 0. \end{aligned}$$

Thus $\phi(z) \in R_\varphi$ and hence $z \in \phi^{-1}(R_\varphi)$. It follows that $|\lambda^{-1}(\{\varphi\})| = |O_G(x)|$ for all φ. By using the argument before the proposition,

$$\begin{aligned} \gamma(E/K) &= |\operatorname{Hom}(E/K, C/K)| \\ &= |O_G(x)|\,|\operatorname{Hom}(F/K, C/K)| \\ &= |O_G(x)|\gamma(F/K). \end{aligned}$$

Since $\gamma(E/F) = \gamma(F(x)/F) = |O_G(x)|$ by Proposition 6.5, the claim follows.

For the general case, we assume that $E = F(x_1, \dots, x_n)$. We proceed by induction on n. If $n = 0$, then $E = F$ and the result is trivial. If $n > 0$, let $L = F[x_1, \dots, x_{n-1}]$ and $E = L(x_n)$. The case proved implies that $\gamma(E/F) = \gamma(E/L)\gamma(L/F)$. By the inductive hypothesis, $\gamma(L/K) = \gamma(L/F)\gamma(F/K)$. Thus

$$\gamma(E/F)\gamma(F/K) = \gamma(E/L)\gamma(L/F)\gamma(F/K) = \gamma(E/L)\gamma(L/K) = \gamma(E/K),$$

again using the previous case. □

§ 7. Separable extensions

7.1. Definition. Let E/K be an extension and $x \in E$ an algebraic element over K. Then x is **separable** over K if x is a simple root of $f(x, K)$.

An algebraic extension E/K is **separable** if every $x \in E$ is separable over K. Then K/K is separable.

7.2. Exercise. Prove that an element x is separable over K if and only if x is a simple root of a polynomial with coefficients in K.

If F/K is a subextension of E/K and $x \in E$ is separable over K, then x is separable over F.

7.3. Exercise. If C is an algebraic closure of K, $x \in C$ and $G = \operatorname{Gal}(C/K)$. Prove that the following statements are equivalent:

1) x is separable over K.
2) Every $y \in O_G(x)$ is separable over K.
3) $\gamma(K(x)/K) = [K(x):K] = \deg f(x, K)$.

Let K be any field and $g \in K[X]$. Let z be a root of g. Then z is a multiple root of g if and only if z is a root of g'.

7.4. Exercise. Prove that if K has characteristic zero or K is finite, then every algebraic extension of K is separable.

7.5. Example. Let $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Then $[E:\mathbb{Q}] = 4$ and $\operatorname{Gal}(E/\mathbb{Q}) \simeq C_2 \times C_2$. The extension $E/\mathbb{Q}$ is normal, as it is the splitting field of $(X^2-2)(X^2-3)$ and it is separable as $\mathbb{Q}$ has characteristic zero.

7.6. Example. Let E be a splitting field of $X^4 - 2$ over $\mathbb{Q}$. Then $E/\mathbb{Q}$ is normal and separable. Note that $E = \mathbb{Q}(\sqrt[4]{2}, i)$, so

$$[E:\mathbb{Q}] = 8 = |\operatorname{Gal}(E/\mathbb{Q})|.$$

Let us compute $\operatorname{Gal}(E/\mathbb{Q})$. If $\sigma \in \operatorname{Gal}(E/\mathbb{Q})$, then $\sigma(\sqrt[4]{2}) \in \{\sqrt[4]{2}, -\sqrt[4]{2}, \sqrt[4]{2}i, -\sqrt[4]{2}i\}$ and $\sigma(i) \in \{-i, i\}$. Two examples are

$$\alpha\colon \begin{cases} \sqrt[4]{2} \mapsto \sqrt[4]{2}i, \\ i \mapsto i, \end{cases} \qquad \beta\colon \begin{cases} \sqrt[4]{2} \mapsto \sqrt[4]{2}, \\ i \mapsto -i. \end{cases}$$

It follows that $\operatorname{Gal}(E/\mathbb{Q})$ is isomorphic to the group $\langle \alpha, \beta \rangle$, which turns out to be isomorphic to the dihedral group of eight elements.

Another consequence: If $E = K(S)$, then E/K is separable if and only if every $x \in S$ is separable over K. One first does the case $E = K(x)$ and then proceeds by induction.

7.7. Exercise. Let $K \subseteq F \subseteq E$ be a tower of fields. Prove that E/K is separable if and only if F/K and E/F are separable.

7.8. Exercise. Let E/K and F/K be extensions. Prove that if F/K is separable, then EF/E is separable.

7.9. Example. Let p be a prime and let t be transcendental over $\mathbb{F}_p = \mathbb{Z}/p$. Let $K = \mathbb{F}_p(t^p)$ and let $E = \mathbb{F}_p(t)$. Then E/K has degree p and is not separable.

Indeed, E/K is algebraic because $f = f(t, K) = X^p - t^p$ (it is an exercise to check that f is irreducible in $K[X]$). Moreover, $f = (X-t)^p \in E[X]$, and so t has multiplicity p in f. It follows that t is not separable over K. Otherwise there is a polynomial $g \in K[X]$ such that t is a simple root of g, and hence, in particular, $g(t) = 0$. However, we then must have $f \mid g$, and thus t cannot be a simple root of g.

Note that, in this case, E/K is normal, as E is the splitting field of f over K.

If E/K is algebraic, then

$$F = \{x \in E : x \text{ is separable over } K\}$$

is a subfield of E that contains K. It is known as the **separable closure** of K with respect to E. Note that $F = K(F)$, as $K(F)$ is separable because it is generated by separable elements. Moreover, F/K is separable and E/F is a **purely inseparable** extension, meaning that for every $x \in E \setminus F$, x is not separable over K.

7.10. Proposition. *If E/K is separable and finite, then $E = K(x)$ for some $x \in E$.*

Proof. Let us assume that K is finite. Then E is finite and hence the multiplicative group $E^\times = E \setminus \{0\}$ is cyclic, say $E^\times = \langle x \rangle$. It follows that $E = K(x)$.

Let us now assume that K is infinite. We first consider the case $E = K(x, y)$. The general case $E = K(x_1, \dots, x_n)$ is left as an exercise, one needs to proceed by induction. Let $n = [E : K]$ and C be an algebraic closure of K containing E. Write $\mathrm{Hom}(E/K, C/K) = \{\sigma_1, \dots, \sigma_n\}$. Let

$$f = \prod_{1 \leq i < j \leq n} ((\sigma_i(y) - \sigma_j(y)) + X(\sigma_i(x) - \sigma_j(x))) \in C[X].$$

Then $f \neq 0$, as f is a product of non-zero polynomials. Since K is infinite, there exists a non-zero $c \in K$ such that $f(c) \neq 0$. For any $r, s \in \{1, \dots, n\}$ with $r \neq s$,

$$\sigma_r(y) - \sigma_s(y) + c(\sigma_r(x) - \sigma_s(x)) \neq 0,$$

as $f(c) \neq 0$. It follows that $\sigma_r(y + cx) \neq \sigma_s(y + cx)$. Thus $\gamma(K(y + cx)/K) \geq n$. Now

$$n \geq [K(y + cx) : K] = \gamma(K(y + cx)/K) \geq n,$$

so $[K(y + cx) : K] = n$ and hence $K(y + cx) = E$. $\square$

For example, $\mathbb{Q}(\sqrt{2}, i) = \mathbb{Q}(\sqrt{2} + i)$.

§ 8. Steinitz' theorem

8.1. Theorem (Steinitz). *Let E/K be a finite extension. Then $E = K(x)$ for some $x \in E$ if and only if E/K admits finitely many subextensions.*

Proof. We may assume that K is infinite; otherwise, the result is trivial. We first prove $\implies$. Let us assume that $E = K(x)$ for some x. We claim that the map

$$\Psi \colon \{F : K \subseteq F \subseteq E\} \to \{g \in E[X] : g \text{ is a monic divisor of } f(x, K)\},$$
$$F \mapsto f(x, F),$$

is injective. Take F_0 such that $K \subseteq F_0 \subseteq F \subseteq E$ and $f(x, F) = f(x, F_0)$. Then

$$[E : F_0] = [F_0(x) : F_0] = \deg f(x, F_0) = m = [F(x) : F] = [E : F]$$

and hence $F = F_0$.

In general, let F_1 and F_2 be such that $K \subseteq F_1, F_2 \subseteq E$ and $f(x, F_1) = f(x, F_2)$. Let $F_0 = F_1 \cap F_2$. Then $f = f(x, F_1) = f(x, F_2) \in F_0[X]$ and hence $f(x, F_0) = f$. Hence we can apply what we proved before to $F_0 \subseteq F_1$ and $F_0 \subseteq F_2$, to obtain that $F_1 = F_0 = F_2$. It follows that Ψ is injective and hence there are finitely many fields between K and E.

Let us prove $\impliedby$. Let us assume that $E = K(x, y)$. For each $a \in K$, we consider the extension $K(ay + x)/K$. By assumption, there exist $a, b \in K$ such that $a \neq b$ and

$$K(x + ay) = K(x + by) = L.$$

We claim that $L = E$. Note that $x + ay \in L$ and $x + by \in L$, so $(a - b)y \in L$ and hence, since $K \subseteq L$, it follows that $y \in L$. Thus $x \in L$ and therefore $L = E$. $\square$

As a consequence, if E/K is finite and separable, then E/K admits finitely many subextensions.

§ 9. Galois extensions

Let E/K be an algebraic extension. Assume that $E = K(S)$ and let C be an algebraic closure of K containing E. Let

$$T = \{y \in C : y \text{ is a root of } f(x,K) \text{ for } x \in S\}$$

and let $L = K(T)$. Then $E \subseteq L$, as $S \subseteq T$. The extension L/K is normal, as L/K is a splitting field of the family $\{f(x,K) : x \in S\}$. Moreover, L is the smallest normal extension of K containing E. The field L is the **normal closure** of E (with respect to C).

9.1. Exercise. If E/K is finite, then L/K is finite

9.2. Exercise. If E/K is separable, then L/K is separable.

Let E/K be an extension and $S \subseteq \mathrm{Gal}(E/K)$ be a subset. The set

$${}^{S}E = \{x \in E : \sigma(x) = x \text{ for all } \sigma \in S\}$$

is a subfield of E that contains K. The subfield ${}^{S}E$ is known as the **fixed field** of S.

9.3. Definition. Let E/K be an algebraic extension and $G = \mathrm{Gal}(E/K)$. Then E/K is a **Galois extension** if ${}^{G}E = K$.

Clearly, K/K is a Galois extension. Note that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not a Galois extension. Why?

9.4. Exercise. Prove that $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ is a Galois extension.

9.5. Exercise. If the characteristic of K is different from two and E/K is an extension of degree 2, then it is Galois.

9.6. Exercise. Let E/K be an algebraic extension and $G = \mathrm{Gal}(E/K)$. Let $F = {}^{G}E$. Prove that $\mathrm{Gal}(E/F) = G$ and hence E/F is a Galois extension.

9.7. Proposition. *Let E/K be an algebraic extension. Then E/K is a Galois extension if and only if E/K is normal and separable.*

Proof. Let $G = \mathrm{Gal}(E/K)$. Let us first assume that E/K is Galois. For $x \in E$ let

$$f_x = \prod_{y \in O_G(x)} (X - y) = \sum a_i X^i \in E[X].$$

If $\varphi \in G$, then

$$\overline{\varphi}(f_x) = \prod_{y \in O_G(x)} (X - \varphi(y)) = f_x,$$

as if $O_G(x) = \{\sigma_1(x), \dots, \sigma_r(x)\}$, then $\varphi(\sigma_i(x)) = (\varphi\sigma_i)(x) = \sigma_j(x)$ for some j. Since

$$\sum a_i X^i = f_x = \overline{\varphi}(f_x) = \sum \varphi(a_i) X^i,$$

it follows that $a_i \in {}^G E = K$ for all i. Thus $f_x \in K[X]$ and E/K is a splitting field of the family $\{f_x : x \in E\}$. In particular, E/K is normal. Moreover, x is a simple root of $f_x \in K[X]$ and hence x is separable over K.

Conversely, let $x \in {}^G E$. Since E/K is normal, then $f(x,K) = \prod_{y \in O_G(x)} (X-y)^m$ for some m. Since E/K is separable, $m = 1$. Moreover $x \in {}^G E$, so $O_G(x) = \{x\}$. Thus $f(x,K) = \prod_{y \in O_G(x)} (X-y) = X - x$ and $x \in K$. □

9.8. Definition. Let K be a field and $f \in K[X]$. Then f is **separable** if all roots of f are simple (in some algebraic closure of K).

9.9. Proposition. *Let E/K be a finite extension. Then E/K is a Galois extension if and only if E is a splitting field over K of a separable polynomial $f \in K[X]$.*

Proof. Let us assume first that E/K is a Galois extension. Since E/K is finite and separable, $E = K(x)$ by Proposition 7.10. Then E/K is a splitting field of $f(x,K)$ since E/K is normal. Since E/K is separable, x is separable over K. Thus x is a simple root of $f(x,K)$ and hence $f(x,K)$ is separable. Conversely, let $x_1, \dots, x_r$ be the roots of a separable polynomial $f \in K[X]$. Then $E = K(x_1, \dots, x_r)$ is separable and normal. □

In the previous case, $\mathrm{Gal}(E/K)$ is known as the **Galois group** of the polynomial f. The notation is $\mathrm{Gal}(f,K)$. If $n = \deg f$ and $x_1, \dots, x_n$ are the roots of f, then any $\varphi \in \mathrm{Gal}(f,K)$ permutes the roots of f, that is φ permutes the set $\{x_1, \dots, x_n\}$. In particular, $\mathrm{Gal}(f,K)$ is isomorphic to a subgroup of $\mathbb{S}_n$ and hence $|\mathrm{Gal}(f,K)|$ divides $n!$.

9.10. Proposition. *Let E/K be a normal extension and F be the separable closure of K with respect to E. Then F/K is a Galois extension.*

Proof. Let C/K be an algebraic closure such that $E \subseteq C$. Let $\sigma \in \mathrm{Hom}(F/K, C/K)$, and let $\varphi \in \mathrm{Hom}(E/K, C/K)$ be such that $\varphi|_F = \sigma$. Since E/K is normal, $\varphi(E) = E$. Let $x \in F$. Then $\sigma(x) = \varphi(x) \in E$. Thus $f(\sigma(x), K) = f(x,K)$ and $\sigma(x)$ is separable over K, which implies that $\sigma(x) \in F$. Thus F/K is normal. Since F/K is separable, it follows that F/K is a Galois extension by Proposition 9.7. □

Some easy facts.

9.11. Exercise. Let E/K be a separable extension and L/K be the normal closure of E in some algebraic closure C that contains E. Prove that L/K is a Galois extension.

9.12. Exercise. Let E/K be a finite extension. Prove that E/K is Galois if and only if $[E:K] = |\mathrm{Gal}(E/K)|$.

For the previous exercise, note that if E/K is a finite extension, then

$$|\mathrm{Gal}(E/K)| \le \gamma(E/K) \le [E:K].$$

The first inequality is equality if and only if E/K is normal. The second inequality is equality if and only if E/K is separable.

9.13. Exercise. Let E/K be a Galois extension and F/K be a subextension of E/K. Prove that E/F is a Galois extension.

9.14. Theorem (Artin). *Let E be a field and G be a finite group of automorphisms of E. If $K = {}^G E$, then E/K is a Galois extension, $[E:K] = |G|$ and* $\operatorname{Gal}(E/K) = G$.

Before proving the theorem, we need a lemma.

9.15. Lemma. *Let E/K be a separable extension such that* $\deg f(x,K) \leq m$ *for all* $x \in E$. *Then E/K is finite and $[E:K] \leq m$.*

Proof. Let $z \in E$ be of maximal degree. If $x \in E$, then $K(x,z)/K$ is separable. There exists y such that $K(x,z) = K(y)$. Then

$$K(z) \subseteq K(x,z) = K(y).$$

Since $\deg f(z,K) \leq \deg f(y,K)$, $\deg f(z,K) = \deg f(y,K)$. Hence $K(y) = K(z)$. In particular, $x \in K(z)$ and therefore $E = K(z)$. $\square$

Now we are ready to prove Artin's theorem:

Proof of Theorem 9.14. Note that $G \subseteq \operatorname{Gal}(E/K)$. Let $x \in E$ and

$$f_x = \prod_{y \in O_G(x)} (X - y).$$

Since $f_x \in K[X]$, the extension E/K is normal and separable (as it is a splitting field of a family of separable polynomials), so E/K is a Galois extension. Moreover,

$$\deg f(x,K) \leq \deg f_x = |O_G(x)| \leq |G|.$$

By the previous lemma, E/K is finite and $[E:K] \leq |G|$. This implies that

$$|\operatorname{Gal}(E/K)| = [E:K] \leq |G|$$

and hence $|\operatorname{Gal}(E/K)| = |G|$. $\square$

9.16. Example. Let $E = K(X,Y)$ and $\sigma \colon K[X,Y] \to E$ be the ring homomorphism given by $\sigma(X) = Y$ and $\sigma(Y) = X$. Note that σ is bijective, as $\sigma^2 = \mathrm{id}$. The map σ induces a field homomorphism $\overline{\sigma} \colon E \to E$ such that $\overline{\sigma}^2 = \mathrm{id}$. Recall that such a homomorphism is given by $f/g \mapsto \sigma(f)/\sigma(g)$. Let $G = \langle \overline{\sigma} \rangle$. Then $|G| = 2$. We claim that ${}^G E = K(X+Y, XY)$. Let $F = K(X+Y, XY)$. We only prove that ${}^G E \subseteq F$, as the other inclusion is trivial. Artin's theorem implies that $[E : {}^G E] = 2$ and $E = F(X)$, as X is a root of the polynomial $Z^2 - (X+Y)Z + XY$. Then $[E:F] \leq 2$ and $[{}^G E : F] = 1$.

§ 10. Galois' correspondence

A **partially order set** (or **poset**) is a pair $(X, \leq)$, where X is a non-empty set and $\leq$ is a reflexive, antisymmetric and transitive relation on X. This means:

1) $x \leq x$ for all $x \in X$,
2) $x \leq y$ and $y \leq x$ imply $x = y$, and
3) $x \leq y$ and $y \leq z$ imply $x \leq z$.

Let $(X, \leq)$ be a partially ordered set and $x, y \in X$. An element z of a poset $(X, \leq)$ is an **upper bound** of x and y if $x \leq z$ and $y \leq z$. And ξ is a **least upper bound** of x and y if it is an upper bound with $\xi \leq z$ for every upper bound z of x and y. Similarly, one defines lower bounds and greatest lower bounds.

10.1. Definition. A **lattice** is a partially ordered set $\mathcal{L}$ in which each pair of elements $x, y \in \mathcal{L}$ has a least upper bound $x \vee y$ and a greatest lower bound $x \wedge y$.

The basic example is the following. Let X be a set and $\mathcal{P}(X)$ be the collection of all subsets of X. The relation $A \leq B \iff A \subseteq B$ turns $\mathcal{P}(X)$ into a lattice with $A \vee B = A \cup B$ and $A \wedge B = A \cap B$.

10.2. Example. Let G be a group and $\mathcal{L}(G)$ be the collection of subgroups of G. The relation

$$H \leq K \iff H \subseteq K$$

turns $\mathcal{L}(G)$ into a lattice with $H \vee K = \langle H, K \rangle$ and $H \wedge K = H \cap K$.

10.3. Example. Let E/K be a field extension and $\mathcal{L}(E/K)$ be the collection of intermediate fields. The relation

$$F \leq L \iff F \subseteq L$$

turns $\mathcal{L}(E/K)$ into a lattice with $F \vee L = FL$ and $F \wedge L = F \cap L$.

10.4. Example. Let $n \in \mathbb{Z}$ be a positive integer and $\mathcal{D}(n)$ be the collection of positive integer divisors $d \in \mathbb{Z}$ of n. The relation

$$l \leq m \iff l \mid m$$

turns $\mathcal{D}(n)$ into a lattice with $l \vee m = \operatorname{lcm}(l, m)$ and $l \wedge m = \gcd(l, m)$.

A map $f \colon \mathcal{L}_1 \to \mathcal{L}_2$ between two lattices is said to be **order-reversing** if $x \leq_1 y$ implies $f(y) \leq_2 f(x)$.

We shall need an exercise.

10.5. Exercise. Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be lattices and $f \colon \mathcal{L}_1 \to \mathcal{L}_2$ be a bijection such that f and its inverse are both order reversing. Then

$$f(x \vee y) = f(x) \wedge f(y), \quad f(x \wedge y) = f(x) \vee f(y)$$

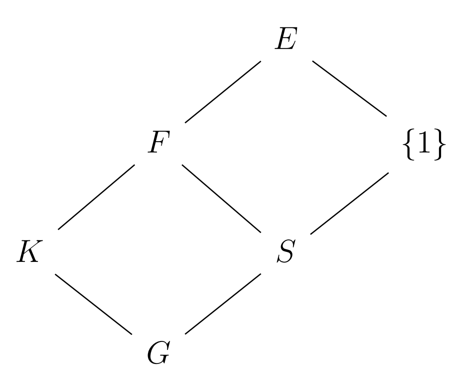
for all $x, y \in \mathcal{L}_1$.

10.6. Theorem (Galois). *Let E/K be a finite Galois extension and $G = \mathrm{Gal}(E/K)$. There exists a bijective correspondence*

$$\mathcal{L}(E/K) = \{F : K \subseteq F \subseteq E \text{ subfields}\} \leftrightarrow \{S : S \text{ is a subgroup of } G\} = \mathcal{L}(G).$$

The correspondence is given by $\alpha \colon F \mapsto \mathrm{Gal}(E/F)$ and $\beta \colon {}^{S}E \leftarrow\!\shortmid S$. Moreover, the following conditions hold:

1) *α and β are order-reversing bijections.*
2) *$[F : K] = (G : \mathrm{Gal}(E/F))$ and $(G : S) = [{}^{S}E : K]$.*
3) *F/K is a Galois extension if and only if $\mathrm{Gal}(E/F)$ is a normal subgroup of G.*



Proof. Let

$$\alpha\colon \mathcal{L}(E/K) \to \mathcal{L}(G), \quad \alpha(F) = \mathrm{Gal}(E/F),$$
$$\beta\colon \mathcal{L}(G) \to \mathcal{L}(E/K), \quad \beta(S) = {}^{S}E.$$

A routine exercise shows that α and β are well-defined. We first note that

$$\beta(\alpha(F)) = \beta(\mathrm{Gal}(E/F)) = {}^{\mathrm{Gal}(E/F)}E = F$$

since E/F is a Galois extension. Moreover,

$$\alpha(\beta(S)) = \alpha({}^{S}E) = \mathrm{Gal}(E/{}^{S}E) = S$$

by Artin's theorem, as S is finite.

It is straightforward to check that α and β are order-reversing bijections.

Let F be a subfield of E containing K and $S = \alpha(F)$. Then

$$[F:K] = \frac{[E:K]}{[E:F]} = \frac{|G|}{|S|} = (G:S).$$

Let C be an algebraic closure of K that contains E. If $S = \mathrm{Gal}(E/F)$, then $F = {}^{S}E$.

We need to prove that F/K is normal if and only if S is normal in G. Let us first prove $\Longrightarrow$. Let $\tau \in S$ and $\sigma \in G$. Since F/K is normal, $\sigma|_F \in \mathrm{Aut}(F)$. Thus $\sigma^{-1}(F) = F$. In particular, if $x \in F$, then $\sigma^{-1}(x) \in F$ and

$$\sigma\tau\sigma^{-1}(x) = \sigma\sigma^{-1}(x) = x.$$

Conversely, let $\varphi \in \mathrm{Hom}(F/K, C/K)$. There exists $\Phi\colon E \to C$ such that $\Phi|_F = \varphi$. Since E/K is normal, $\Phi(E) = E$ and hence $\Phi \in G$. We claim that $\varphi(x) \in F$ for all $x \in F$. Note that $F = {}^{S}E$, so

$$\tau\varphi(x) = \tau\Phi(x) = \Phi\Phi^{-1}\tau\Phi(x) = \Phi(x) = \varphi(x)$$

for all $\tau \in S$, as $\Phi^{-1}\tau\Phi \in S$. This means that $\varphi(x) \in {}^{S}E = F$.

Let us compute $\mathrm{Gal}(F/K)$. Since F/K is normal, the map $\lambda\colon G \to \mathrm{Gal}(F/K)$, $\sigma \mapsto \sigma|_F$, is a surjective group homomorphism such that $\ker\lambda = S$. The first isomorphism theorem implies that $\mathrm{Gal}(F/K) \simeq G/S$. □

Some easy consequences.

10.7. Exercise. If E/K is a Galois extension of degree n and p is a prime number dividing n, then E/K admits a subextension of degree n/p.

10.8. Exercise. If E/K is a Galois extension of degree $p^{\alpha}m$ with p a prime number coprime with m, then E/K admits a subextension of degree m.

10.9. Definition. An extension E/K is **abelian** if E/K is a Galois extension with $\mathrm{Gal}(E/K)$ abelian.

10.10. Exercise. If E/K is an abelian extension of degree n and d divides n, then E/K admits a subextension of degree d.

10.11. Definition. An extension E/K is **cyclic** if E/K is a Galois extension with $\mathrm{Gal}(E/K)$ cyclic.

10.12. Example. The extension $\mathbb{Q}(\sqrt{2},\sqrt{3})/\mathbb{Q}$ admits exactly three non-trivial subextensions:

$$\mathbb{Q}(\sqrt{2})/\mathbb{Q},\quad \mathbb{Q}(\sqrt{3})/\mathbb{Q},\quad \mathbb{Q}(\sqrt{6})/\mathbb{Q},$$

as $\operatorname{Gal}(\mathbb{Q}(\sqrt{2},\sqrt{3})/\mathbb{Q})\simeq C_2\times C_2$.

10.13. Example. Let $\omega\in\mathbb{C}\setminus\{1\}$ be such that $\omega^5=1$. Then

$$f(\omega,\mathbb{Q})=1+X+X^2+X^3+X^4$$

and $\mathbb{Q}(\omega)/\mathbb{Q}$ has degree four. Moreover, $\mathbb{Q}(\omega)/\mathbb{Q}$ is a Galois extension and

$$\operatorname{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})\simeq C_4.$$

If $\sigma\in\operatorname{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})$, then $\sigma(\omega)=\omega^i$ for some $i\in\{1,\dots,4\}$. Moreover, each map $\omega\mapsto\omega^i$, for $i\in\{1,\dots,4\}$, induces an automorphism of $\mathbb{Q}(\omega)/\mathbb{Q}$. Thus $|\operatorname{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})|=4$. Now

$$\sigma_i^k=\mathrm{id}\iff\omega^{i^k}=\sigma_i^k(\omega)=\omega\iff i^k\equiv 1 \bmod 5.$$

Thus the map σ_2 given by $\omega\mapsto\omega^2$ has order four.

Since $\operatorname{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})=\langle\sigma\rangle$, where $\sigma(\omega)=\omega^2$, is cyclic of order four, the extension $\mathbb{Q}(\omega)/\mathbb{Q}$ has a unique degree-two subtextension $F/\mathbb{Q}$. Note that $|\langle\sigma^2\rangle|=2$ and $\sigma^2(\omega)=\omega^4=\omega^{-1}$. Thus $F={}^{\langle\sigma^2\rangle}\mathbb{Q}(\omega)$. Let $\theta=\omega+\omega^{-1}$. Then

$$\theta^2=\omega^2+\omega^3+2=-(1+\omega+\omega^{-1})+2=1-\theta$$

and hence θ is a root of X^2+X-1. It follows that

$$\theta\in\{(-1+\sqrt{5})/2,(-1-\sqrt{5})/2\}.$$

Therefore $F=\mathbb{Q}(\sqrt{5})$.

Let us mention some other consequences (of the fact that the correspondence depends on order-reversing bijections).

10.14. Exercise. Let E/K be a finite Galois extension and $G=\operatorname{Gal}(E/K)$. If S and T are subgroups of G, then ${}^{\langle S,T\rangle}E={}^{S}E\cap{}^{T}E$ and ${}^{S\cap T}E={}^{S}E\,{}^{T}E$.

10.15. Exercise. Let E/K be a finite Galois extension and $F,L\in\mathcal{L}(E/K)$. Prove that $\operatorname{Gal}(E/FL)=\operatorname{Gal}(E/F)\cap\operatorname{Gal}(E/L)$ and $\operatorname{Gal}(E/F\cap L)=\langle\operatorname{Gal}(E/F),\operatorname{Gal}(E/L)\rangle$.

10.16. Exercise. Let E/K be a finite Galois extension and $G=\operatorname{Gal}(E/K)$. Assume that G is the direct product $G=S\times T$ of the groups S and T. Let $F={}^{S}E$ and $L={}^{T}E$. Then $F\cap L=K$ and $FL=E$.

10.17. Proposition. *Let $E_1/K,E_2/K$ be Galois extensions. If $E=E_1E_2$, then E/K is a Galois extension. If, moreover, E_1/K and E_2/K are finite, then*

$$\theta\colon \operatorname{Gal}(E/K)\to\operatorname{Gal}(E_1/K)\times\operatorname{Gal}(E_2/K),\quad \sigma\mapsto(\sigma|_{E_1},\sigma|_{E_2}),$$

is an injective group homomorphism.

Proof. Since E_1/K is algebraic, E_1E_2/E_2 is algebraic. Since E_2/K is algebraic, E_1E_2/K is algebraic. Similarly, E_1E_2/K is separable.

Let C/K be an algebraic closure such that $E_1E_2 \subseteq C$. If $\sigma \in \operatorname{Hom}(E_1E_2/K, C/K)$, then $\sigma(E_1E_2) \subseteq \sigma(E_1)\sigma(E_2) = E_1E_2$ (do this calculation as an exercise using the fact that E_1/K and E_2/K are normal extensions). Thus E_1E_2/K is normal.

If both E_1/K and E_2/K are finite, then E_1E_2/K is finite.

Then θ is a group homomorphism. We claim that the map θ is injective. Let $\sigma \in \ker\theta$. Then $\sigma|_{E_i} = \mathrm{id}_{E_i}$ for all $i \in \{1,2\}$. Let $S = \langle\sigma\rangle \subseteq \operatorname{Gal}(E/K)$ and $F = {}^{S}E$. Then $E_i \subseteq F$ for all $i \in \{1,2\}$ and hence $E \subseteq F$. It follows that $F = E = {}^{\{\mathrm{id}\}}E$ and therefore $S = \{\mathrm{id}\}$, so $\sigma = \mathrm{id}$. □

10.18. Exercise. Let $E_1/K, \dots, E_r/K$ be finite Galois extensions such that for each j one has $E_j \cap (E_1 \cdots E_{j-1}E_{j+1} \cdots E_r) = K$. If $E = E_1E_2 \cdots E_r$, then

$$\operatorname{Gal}(E/K) \simeq \operatorname{Gal}(E_1/K) \times \cdots \times \operatorname{Gal}(E_r/K).$$

In this case, $[E : K] = \prod_{i=1}^{r}[E_i : K]$.

§ 11. The fundamental theorem of algebra

We now present an easy proof of the fundamental theorem of algebra based on the ideas of Galois Theory. We need the following well-known facts:

1) Every real polynomial of odd degree admits a real root. This means that $\mathbb{R}$ does not admit extensions of odd degree > 1.

2) Every complex number admits a square root in $\mathbb{C}$. This means that $\mathbb{C}$ does not admit degree-two extensions.

11.1. Theorem. *The field $\mathbb{C}$ is algebraically closed.*

Proof. Let $E/\mathbb{C}$ be an algebraic finite extension. Then $E/\mathbb{R}$ is a finite separable extension of even degree. There exists a Galois extension $L/\mathbb{R}$ such that $E \subseteq L$, so $[L : \mathbb{R}]$ is even. Let $G = \operatorname{Gal}(L/\mathbb{R})$. Then $|G| = 2^m s$ for some odd number s. If T is a 2-Sylow subgroup of G, then there exists a subextension $F/\mathbb{R}$ of degree s. Since $\mathbb{R}$ does not admit extensions of odd degree > 1, $s = 1$ and hence G is a 2-group. Since $L/\mathbb{R}$ is a Galois extension, $L/\mathbb{C}$ is a Galois extension. In particular, $|\operatorname{Gal}(L/\mathbb{C})| = 2^{m-1}$. If $m > 1$, let U be a subgroup of $\operatorname{Gal}(L/\mathbb{C})$ of order 2^{m-2}. Then U corresponds to a subextension $L_1/\mathbb{C}$ of degree two, a contradiction. Hence $m = 1$ and $[L : \mathbb{C}] = 1$, so $L = \mathbb{C}$ and $E = \mathbb{C}$. □

§ 12. Purely inseparable extensions

Let E/K be an algebraic extension. In Chapter 7 we defined the **separable closure** of K with respect to E as the field

$$F = \{x \in E : x \text{ is separable over } K\}.$$

Note that $K \subseteq F \subseteq E$ and $F = K(F)$. Moreover, F/K is separable and E/F is a **purely inseparable** extension, meaning that for every $x \in E \setminus F$, the polynomial $f(x, F)$ is not separable.

The number $[E : K]_{\mathrm{ins}} = [E : F]$ is known as the **degree of inseparability** of E/K. Clearly, E/K is separable if and only if $[E : K]_{\mathrm{ins}} = 1$ and E/K is purely inseparable if and only if $[E : K]_{\mathrm{ins}} = [E : K]$.

12.1. Exercise. Let K be a field of characteristic $p>0$ and $f\in K[X]$ be irreducible. If f is not separable, then $f=g(X^p)$ for some $g\in K[X]$.

12.2. Proposition. *Let K be a field of characteristic $p>0$ and E/K be an algebraic extension. The following statements are equivalent:*

1) *E/K is purely inseparable.*
2) *If $x\in E$, then $x^{p^m}\in K$ for some $m\geq 0$.*
3) *If $x\in E$, then $f(x,K)=X^{p^m}-a$ for some $a\in K$ and $m\geq 0$.*
4) *$\gamma(E/K)=1$.*

Proof. We first prove 1) $\implies$ 2). Let $x\in E$ and $f=f(x,K)$. Assume x is not separable. Then $f(x)=0$ and $f'(x)=0$, as x is not a simple root. By Exercise 12.1, $f=g(X^p)$ for some $g\in K[X]$. We now proceed by induction on the degree of x. The result is true for elements of degree one. So assume the result holds for the element of degree $\leq n$ for some $n\geq 1$. If $x\in E$ is such that $\deg f(x,K)=n+1$, then, since $f(x,K)=g(X^p)$, the element x^p has degree $\leq n$. By the inductive hypothesis, $x^{p^{m+1}}=(x^p)^{p^m}\in K$.

We now prove 2) $\implies$ 3). Let $x\in E$ and m be the minimal positive integer such that $x^{p^m}\in K$. Then x is a root of $X^{p^m}-x^{p^m}\in K[X]$. Since $X^{p^m}-x^{p^m}=(X-x)^{p^m}$, it follows that

$$f(x,K)=(X-x)^r=X^r+\cdots+(-1)^r x^r$$

for some $r\in\{1,\dots,p^m\}$. Write $r=p^s t$ for some integer t coprime with p and s such that $0\leq s\leq m$. Let $a,b\in\mathbb{Z}$ be such that $ar+bp^m=p^s$. Then

$$x^{p^s}=x^{ar+bp^m}=(x^r)^a\left(x^{p^m}\right)^b\in K.$$

The minimality of m implies that $s\geq m$ and hence $s=m$. Now $p^m t=p^s t=r\leq p^m$, so $t=1$. This means $f(x,K)=X^{p^m}-x^{p^m}$.

We now prove 3) $\implies$ 4). Let C/K be an algebraic closure of K containing E and $x\in E$. Let $\sigma\in\operatorname{Hom}(E/K,C/K)$. We claim that $\sigma(x)=x$. Since $f(x,K)=X^{p^m}-a$,

$$(\sigma(x))^{p^m}=\sigma\left(x^{p^m}\right)=\sigma(a)=a=x^{p^m}.$$

It follows that $\sigma(x)$ is a root of $X^{p^m}-x^{p^m}=(X-x)^{p^m}$. Thus $\sigma(x)=x$.

Finally, we prove that 4) $\implies$ 1). Let C be an algebraic closure of K containing E. Then $\operatorname{Gal}(E/K)\subseteq\operatorname{Hom}(E/K,C/K)=\{\mathrm{id}\}$, as $\gamma(E/K)=1$. If $x\in E$ is separable over K, then

$$f(x,K)=\prod_{y\in O_{\operatorname{Gal}(E/K)}(x)}(X-y)=X-x\in K[X].$$

Thus $x\in K$ and hence E/K is purely inseparable. □

Some consequences:

12.3. Exercise. Let K be a field of characteristic $p>0$ and E/K be finite and purely inseparable. Then $[E:K]=p^s$ for some prime number p and some integer s. Moreover, for any $x\in E$, we have that $x^{[E:K]}\in K$.

For the first part of the previous exercise, write $E=K(x_1,\dots,x_n)$ and proceed by induction on n.

12.4. Exercise. Let K be of characteristic $p > 0$ and E/K be a finite extension such that $[E : K]$ is not divisible by p. Then E/K is separable.

Let K be of characteristic $p > 0$, E/K be finite and F be the separable closure of K in E. Since

$$\gamma(E/K) = \gamma(E/F)\gamma(F/K) = \gamma(F/K),$$

it follows that

$$[E : K] = [E : F]\gamma(E/K) = [E : K]_{\text{ins}}\gamma(E/K).$$

§ 13. Norm and trace

13.1. Definition. Let E/K be a finite extension and C/K be an algebraic closure that contains E. Let $A = \operatorname{Hom}(E/K, C/K)$. For $x \in E$ we define the **trace** of x in E/K as

$$\operatorname{trace}_{E/K}(x) = [E : K]_{\text{ins}} \sum_{\varphi \in A} \varphi(x)$$

and the **norm** of x in E/K as

$$\operatorname{norm}_{E/K}(x) = \left(\prod_{\varphi \in A} \varphi(x) \right)^{[E:K]_{\text{ins}}}.$$

As an optional exercise, one can show that these definitions do not depend on the algebraic closure.

We collect some basic properties as an exercise:

13.2. Exercise. Let E/K be a finite extension. The following statements hold:

1) If E/K is not separable, then $\operatorname{trace}_{E/K}(x) = 0$ for all $x \in E$.
2) If $x \in K$, then $\operatorname{trace}_{E/K}(x) = [E : K]x$.
3) $\operatorname{trace}_{E/K}(x) \in K$ for all $x \in E$.
4) $\operatorname{norm}_{E/K}(x) = 0$ if and only if $x = 0$.
5) If $x \in K$, then $\operatorname{norm}_{E/K}(x) = x^{[E:K]}$.
6) $\operatorname{norm}_{E/K}(x) \in K$ for all $x \in E$.

One proves, moreover, that $\operatorname{trace}_{E/K} \colon E \to K$ satisfies

$$\operatorname{trace}_{E/K}(x + \lambda y) = \operatorname{trace}_{E/K}(x) + \lambda \operatorname{trace}_{E/K}(y)$$

for all $x, y \in E$ and $\lambda \in K$, that is to say that $\operatorname{trace}_{E/K} \colon E \to K$ is a linear form in E. Furthermore $\operatorname{norm}_{E/K} \colon E^\times \to K^\times$ is a group homomorphism.

13.3. Exercise. Let E/K be a finite extension and $x \in E$ be algebraic. If

$$f(x, K) = X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0,$$

then $\operatorname{norm}_{E/K}(x) = ((-1)^n a_0)^{[E:K(x)]}$ and $\operatorname{trace}_{E/K}(x) = -[E : K(x)]a_{n-1}$.

13.4. Example. Let $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Then

$$\operatorname{trace}_{E/\mathbb{Q}}(\sqrt{2}) = 0, \qquad \operatorname{norm}_{E/\mathbb{Q}}(\sqrt{2}) = 4,$$
$$\operatorname{trace}_{E/\mathbb{Q}(\sqrt{2})}(\sqrt{2}) = 2\sqrt{2}, \qquad \operatorname{norm}_{E/\mathbb{Q}(\sqrt{2})}(\sqrt{2}) = 2.$$

13.5. Example. If E/K is a finite Galois extension, then

$$\operatorname{trace}_{E/K}(x) = \sum_{\sigma \in \operatorname{Gal}(E/K)} \sigma(x) \quad \text{and} \quad \operatorname{norm}_{E/K}(x) = \prod_{\sigma \in \operatorname{Gal}(E/K)} \sigma(x)$$

for all $x \in E$. In particular, since $E = K(y)$ for some y by Proposition 7.10,

$$\operatorname{trace}_{E/K}(y) = -a_{n-1} \quad \text{and} \quad \operatorname{norm}_{E/K}(y) = (-1)^n a_0,$$

where $f(y, K) = X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0$.

§ 14. Finite fields

In this section, p will be a prime number.

14.1. Proposition. *Let m be a positive integer. Up to isomorphism, there exists a unique field F_{p^m} of size p^m.*

Proof. Let C be an algebraic closure of the field $\mathbb{Z}/p$ and let $F_{p^m} = \{x \in C : x^{p^m} = x\}$ be the set of roots of $X^{p^m} - X$. Since the polynomial $X^{p^m} - X$ has no multiple roots, $|F_{p^m}| = p^m$. Moreover, F_{p^m} is the unique subfield of C of size p^m.

To prove the uniqueness, it is enough to note that if K is a field of p^m elements, then K is the splitting field of $X^{p^m} - X$ over $\mathbb{Z}/p$. □

Let $K = \mathbb{Z}/p$ and C be an algebraic closure of K. We claim that $C = \bigcup_k F_{p^k}$. If $x \in C$, then x is algebraic over K. Since $K(x)/K$ is finite, $K(x)$ is a finite field, say $|K(x)| = p^r$ for some r. Then $x^{p^r} = x$ and hence $x \in F_{p^r}$.

14.2. Exercise. Prove the following statements:

1) If $x \in F_{p^r}$, then $x^{p^{rk}} = x$ for all $k \geq 0$.
2) $F_{p^m} \subseteq F_{p^n}$ if and only if $m \mid n$.
3) $F_{p^m} \cap F_{p^n} = F_{p^{\gcd(m,n)}}$.

14.3. Proposition. *Every finite extension of a finite field is cyclic.*

Proof. Let $K = \mathbb{Z}/p$. It is enough to show that F_{p^n}/F_{p^m} is cyclic if m divides n.

We first prove that F_{p^n}/K is cyclic. Let

$$\sigma \colon F_{p^n} \to F_{p^n}, \quad x \mapsto x^p.$$

Then $\sigma \in \operatorname{Gal}(F_{p^n}/K)$ (it is bijective because all field homomorphisms are injective and F_{p^n} is finite).

Note that F_{p^n}/K is a Galois extension, as F_{p^n} is the splitting field over K of the separable polynomial $X^{p^n} - X \in K[X]$. Thus $|\operatorname{Gal}(F_{p^n}/K)| = [F_{p^n} : K] = n$.

We claim that σ generates $\operatorname{Gal}(F_{p^n}/K)$. Since $\sigma^i(x) = x^{p^i}$ for all $i \geq 0$, in particular,

$$\sigma^n(x) = x^{p^n} = x.$$

Thus $\sigma^n = \mathrm{id}$ and hence $|\sigma|$ divides n. Let $s = |\sigma|$. We know that $F_{p^n}^{\times} = F_{p^n} \setminus \{0\}$ is cyclic, say $F_{p^n}^{\times} = \langle g \rangle$. Since $|g| = p^n - 1$,

$$g = \sigma^s(g) = g^{p^s}$$

and hence $p^s \equiv 1 \bmod (p^n - 1)$. Thus $p^n - 1$ divides $p^s - 1$ and hence n divides s. Therefore $n = s$ and $\mathrm{Gal}(F_{p^n}/K) = \langle \sigma \rangle$.

For the general case, note that if m divides n, then the Galois group $\mathrm{Gal}(F_{p^n}/F_{p^m})$ is a subgroup of $\mathrm{Gal}(F_{p^n}/K)$. Since $\mathrm{Gal}(F_{p^n}/K)$ is cyclic, the claim follows. □

If $K = \mathbb{Z}/p$ and m divides n, the subextension F_{p^m} corresponds to the unique subgroup of index m of $\mathrm{Gal}(F_{p^n}/K) = \langle \sigma \rangle$. This subgroup is $\langle \sigma^m \rangle$, where for all $x \in F_{p^n}$,

$$\sigma^m(x) = x^{p^m} = x^{|F_{p^m}|}.$$

Note that $\mathrm{Gal}(F_{p^n}/F_{p^m}) = \langle \sigma^m \rangle$. The map σ^m is known as the **Frobenius automorphism**.

14.4. Exercise. Let E/K be an extension of finite fields. Then E/K is cyclic. Moreover, $\mathrm{Gal}(E/K) = \langle \tau \rangle$, where $\tau(x) = x^{|K|}$.

§ 15. Cyclotomic extensions

For $n \geq 1$ let $G_n(K) = \{x \in K : x^n = 1\}$ be the set of n-roots of one in K. Note that $G_n(K)$ is a cyclic subgroup of $K^{\times}$ and that $|G_n(K)|$ divides n.

15.1. Example. $G_n(\mathbb{R}) = \{-1, 1\}$ if n is even and $G_n(\mathbb{R}) = \{1\}$ if n is odd.

15.2. Exercise. Let K be a field of characteristic $p > 0$. Let $n = p^s m$ for some m not divisible by p. Then $G_n(K) = G_m(K)$.

15.3. Exercise. Let q be a prime number. Then $G_n(\mathbb{Z}/q) \simeq \mathbb{Z}/\gcd(n, q-1)$.

Similarly, one can prove that if K is a finite field, then $G_n(K)$ is a cyclic group of order $\gcd(n, |K^{\times}|)$.

15.4. Example. If C is algebraically closed of characteristic coprime with n, then $G_n(C)$ is cyclic of order n, as $X^n - 1$ has all its roots in C and does not contain multiple roots.

Let K be an algebraically closed field and n be such that n is coprime with the characteristic of K. The set of **primitive n-roots** is defined as

$$H_n(K) = \{x \in G_n(K) : |x| = n\}.$$

15.5. Definition. Let K be an algebraically closed field and n be such that n is coprime with the characteristic of K. The **n-th cyclotomic polynomial** is defined as

$$\Phi_n = \prod_{x \in H_n(K)} (X - x) \in K[X].$$

For $n \geq 1$ the Euler's function is defined as

$$\varphi(n) = |\{k : 1 \leq k \leq n,\ \gcd(k, n) = 1\}|.$$

For example, $\varphi(4) = 2$, $\varphi(8) = \varphi(10) = 4$ and $\varphi(p) = p - 1$ for every prime p.

15.6. Proposition. *Let K be an algebraically closed field and n be such that n is coprime with the characteristic of K. Let A be the prime subring of K.*

1) $\deg \Phi_n = \varphi(n)$.
2) $\Phi_n \in A[X]$.

Proof. The first statement is clear. Let us prove 2) by induction on n. The case $n = 1$ is trivial, as $\Phi_1 = X - 1$. Assume that $\Phi_d \in A[X]$ for all d such that $d < n$. In particular,

$$\gamma = \prod_{\substack{d \mid n \\ d \neq n}} \Phi_d \in A[X].$$

Since γ is monic, it follows that $\frac{X^n-1}{\gamma} \in A[X]$. Now the claim follows from

$$X^n - 1 = \prod_{d \mid n} \Phi_d = \Phi_n \prod_{\substack{d \mid n \\ d \neq n}} \Phi_d = \Phi_n \gamma. \qquad \square$$

By taking degree in the equality $X^n - 1 = \prod_{d \mid n} \Phi_d$ one gets

$$n = \sum_{d \mid n} \varphi(d).$$

15.7. Definition. Let $n \geq 2$ and K be a field of characteristic coprime with n. A **cyclotomic extension** of K of index n is a splitting field of $X^n - 1$ over K.

Let C be an algebraic closure of K and $n \geq 2$ be coprime with the characteristic of K. It follows from Definition 15.7 that a cyclotomic extension of K of index n is of the form $K(\omega)/K$ for some $\omega \in H_n(C)$.

15.8. Proposition. *A cyclotomic extension of index n is abelian and of degree a divisor of $\varphi(n)$.*

Proof. Let C be an algebraic closure of K and $n \geq 2$ be coprime with the characteristic of K. Let $\omega \in H_n(C)$ and $K(\omega)/K$ be a cyclotomic extension. Then $K(\omega)/K$ is a Galois extension, as it is a splitting field of a separable polynomial. Let $U = \mathcal{U}(\mathbb{Z}/n)$ be the group of units of $\mathbb{Z}/n$ and

$$\lambda \colon \operatorname{Gal}(K(\omega)/K) \to U, \quad \sigma \mapsto m_\sigma,$$

where m_σ is such that $\sigma(\omega) = \omega^{m_\sigma}$. The map λ is well-defined and it is a group homomorphism, as if $\sigma, \tau \in \operatorname{Gal}(K(\omega)/K)$, then, since

$$(\tau\sigma)(\omega) = \tau(\sigma(\omega)) = \tau(\omega^{m_\sigma}) = \tau(\omega)^{m_\sigma} = (\omega^{m_\tau})^{m_\sigma} = \omega^{m_\sigma m_\tau},$$

it follows that $\lambda(\sigma)\lambda(\tau) = \lambda(\sigma\tau)$. Since λ is injective, $\operatorname{Gal}(K(\omega)/K)$ is isomorphic to a subgroup of the abelian group U. Hence $\operatorname{Gal}(K(\omega)/K)$ is abelian. Moreover,

$$[K(\omega) : K] = |\operatorname{Gal}(K(\omega)/K)|$$

is a divisor of $|U| = \varphi(n)$. $\square$

15.9. Exercise. Prove that a cyclotomic extension $K(\omega)/K$ has degree $\varphi(n)$ if and only if Φ_n is irreducible over K.

Note that Φ_n is irreducible over $\mathbb{Q}$. Some concrete examples:

$$\Phi_1 = X - 1, \quad \Phi_2 = X + 1, \quad \Phi_3 = X^2 + X + 1, \quad \Phi_6 = X^2 - X + 1.$$

If p is a prime number, then $\Phi_p = X^{p-1} + \cdots + X + 1$.

15.10. Example. Φ_5 is irreducible over $\mathbb{Z}/2$. First note that $\Phi_5 = X^4 + \cdots + X + 1$ does not have roots in $\mathbb{Z}/2$. If Φ_5 is reducible, then, since $X^2 + X + 1$ is the unique degree-two monic irreducible polynomial over $\mathbb{Z}/2$, it follows that

$$\Phi_5 = (X^2 + X + 1)(X^2 + X + 1) = (X^2 + X + 1)^2 = X^4 + X^2 + 1,$$

a contradiction.

15.11. Exercise. Prove that $\Phi_{12} = X^4 - X^2 + 1$ is not irreducible over $\mathbb{Z}/5$.

§ 16. Hilbert's theorem 90

16.1. Theorem (Hilbert). *Let E/K be a cyclic extension. Assume that $\operatorname{Gal}(E/K)$ is generated by τ. For $a \in E$, $\operatorname{norm}_{E/K}(a) = 1$ if and only if $a = b/\tau(b)$ for some $b \in E \setminus \{0\}$.*

Proof. Let $n = |\operatorname{Gal}(E/K)|$. We first prove $\Longleftarrow$. If $a = b/\tau(b)$ and $b \neq 0$, then

$$\operatorname{norm}_{E/K}(a) = a\tau(a)\tau^2(a)\cdots\tau^{n-1}(a) = \frac{b}{\tau(b)}\frac{\tau(b)}{\tau^2(b)}\cdots\frac{\tau^{n-1}(b)}{\tau^n(b)} = 1.$$

Now we prove $\Longrightarrow$. Let $a \in E$ be such that $\operatorname{norm}_{E/K}(a) = 1$. For $c \in E$ let

$$\begin{aligned}
d_0 &= ac,\\
d_1 &= a\tau(a)\tau(c),\\
d_2 &= a\tau(a)\tau^2(a)\tau^2(c),\\
&\vdots\\
d_{n-1} &= \underbrace{a\tau(a)\cdots\tau^{n-1}(a)}_{=\operatorname{norm}_{E/K}(a)}\tau^{n-1}(c) = \tau^{n-1}(c).
\end{aligned}$$

Then

$$a\tau(d_j) = a\tau(a)\cdots\tau^{j+1}(a)\tau^{j+1}(c) = d_{j+1}$$

for all $j \in \{0, \ldots, n-2\}$. Let $b = d_0 + \cdots + d_{n-1}$. We claim that $b \neq 0$ for some c. Suppose this is not true, say $b = 0$ for all c. Then

$$0 = ac + (a\tau(a))\tau(c) + \cdots + (a\tau(a)\cdots\tau^{n-1}(a))\tau^{n-1}(c)$$

for every $c \in E$. This implies that $a = 0$ by Dedekind's theorem, a contradiction.

So let $c \in E$ be such that $b \neq 0$. Then

$$\begin{aligned}
\tau(b) &= \tau(d_0) + \cdots + \tau(d_{n-1})\\
&= \tau(ac) + \tau(a\tau(a)\tau(c)) + \cdots + \tau(\tau^{n-1}(c))\\
&= \frac{1}{a}(d_1 + \cdots + d_{n-1}) + \tau^n(c)\\
&= \frac{1}{a}(d_0 + \cdots + d_{n-1})\\
&= b/a.
\end{aligned}$$

$\square$

16.2. Exercise. Let E/K be a cyclic extension. Assume that $\mathrm{Gal}(E/K)$ is generated by τ. Prove that for $a \in E$, $\mathrm{trace}_{E/K}(a) = 0$ if an only if $a = b - \tau(b)$ for some $b \in E \setminus \{0\}$.

16.3. Corollary. *Let $a, b, c \in \mathbb{Z}$ be such that $a^2 + b^2 = c^2$. Then*

$$(a, b, c) = \begin{cases} \lambda(r^2 - s^2, 2rs, r^2 + s^2) & \text{or} \\ \lambda(2rs, r^2 - s^2, r^2 + s^2) \end{cases}$$

for some $r, s \in \mathbb{Z}$ and some $\lambda \in \mathbb{Z}$.

Proof. We work with the extension $\mathbb{Q}(i)/\mathbb{Q}$. Note that $\mathrm{Gal}(\mathbb{Q}(i), \mathbb{Q}) = \{\mathrm{id}, \gamma\}$ is cyclic, where $\gamma \colon \mathbb{Q}(i) \to \mathbb{Q}(i)$, $z \mapsto \overline{z}$, is the complex conjugation. We may assume that $c \neq 0$, otherwise $a = b = 0$ and the result is trivial. Write $(a/c)^2 + (b/c)^2 = 1$ and let

$$\alpha = (a/c) + (b/c)i \in \mathbb{Q}(i).$$

Then $\mathrm{norm}_{\mathbb{Q}(i)/\mathbb{Q}}(\alpha) = 1$. By Hilbert's theorem, there exists $\beta \in \mathbb{Q}(i) \setminus \{0\}$ such that

$$\alpha = a + bi = \frac{\beta}{\gamma(\beta)}.$$

Note that if $m \in \mathbb{Z} \setminus \{0\}$, then $\frac{m\beta}{\gamma(m\beta)} = \frac{\beta}{\gamma(\beta)}$. There exists $m \in \mathbb{Z} \setminus \{0\}$ such that $m\beta \in \mathbb{Z}[i]$, say $m\beta = r + is$ with $r, s \in \mathbb{Z}$ coprime. Then

$$\alpha = \frac{\beta}{\gamma(\beta)} = \frac{m\beta}{\gamma(m\beta)} = \frac{r + is}{r - is} = \frac{r^2 - s^2 + 2rsi}{r^2 + s^2} = \frac{r^2 - s^2}{r^2 + s^2} + \frac{2rs}{r^2 + s^2} i.$$

Note that

$$\gcd(r^2 - s^2, r^2 + s^2) = \gcd(r^2 - s^2, (r^2 - s^2) + (r^2 + s^2)) = \gcd(r^2 - s^2, 2r^2) \leq 2.$$

If the gcd is 1, then

$$(a, b, c) = \lambda(r^2 - s^2, 2rs, r^2 + s^2).$$

Otherwise

$$(a, b, c) = \lambda\left(\frac{r^2 - s^2}{2}, \frac{2rs}{2}, \frac{r^2 + s^2}{2}\right).$$

However, as r and s are coprime such that $r^2 - s^2 \equiv 0 \pmod 2$, they must both be odd, and we can choose coprime integers u and w such that

$$r = u + w,$$
$$s = u - w.$$

Substituting these in, we get

$$\frac{r^2 - s^2}{2} = 2uw, \quad \frac{2rs}{2} = u^2 - w^2, \quad \frac{r^2 + s^2}{2} = u^2 + w^2,$$

giving us the second triple in the statement. □

16.4. Exercise. Let $A, B \in \mathbb{Z}$ be such that $A^2 - 4B$ is not a square. Prove that a solution $(x, y, z) \in \mathbb{Z}^3$ of $x^2 + Axy + By^2 = z^2$ is proportional to

$$(r^2 - Bs^2, 2rs + As^2, r^2 + Ars + Bs^2).$$

16.5. Proposition. *Let $n \geq 2$ and K be a field containing a primitive n-root of one. If $a \in K^\times$ and E/K is a splitting field of $f = X^n - a$, then E/K is cyclic of degree d, where d divides n. Moreover,*

$$d = \min\{k : a^k \in K^n\},$$

where $K^n = \{x \in K : x = y^n \text{ for some } y \in K\}$. Conversely, if E/K is cyclic of degree n, then E/K is a splitting field of an irreducible polynomial of the form $X^n - a$ for some $a \in K^\times$.

Proof. A splitting field of f over K is of the form $K(\alpha)$, where $\alpha^n = a$. Thus $K(\alpha)/K$ is a Galois extension. If $\sigma \in \mathrm{Gal}(K(\alpha)/K)$, then $\sigma(\alpha)$ is a root of f, so $\sigma(\alpha) = \omega_\sigma \alpha$, where $\omega_\sigma \in G_n(K)$. This means that there exists an injective map

$$\lambda\colon \mathrm{Gal}(K(\alpha)/K) \to G_n(K), \quad \sigma \mapsto \omega_\sigma.$$

Moreover, λ is a group homomorphism, as

$$\sigma\tau(\alpha) = \sigma(\tau(\alpha)) = \sigma(\omega_\tau\alpha) = \omega_\tau\sigma(\alpha) = \omega_\tau\omega_\sigma\alpha.$$

Therefore $\mathrm{Gal}(K(\alpha)/K)$ is isomorphic to a subgroup of $G_n(K)$. In particular, $\mathrm{Gal}(K(\alpha)/K)$ is cyclic and $|\mathrm{Gal}(K(\alpha)/K)|$ divides n.

Let $d = |\mathrm{Gal}(K(\alpha)/K)|$. Since $a = \alpha^n$,

$$\mathrm{norm}_{K(\alpha)/K}(\alpha)^n = \mathrm{norm}_{K(\alpha)/K}(a) = a^d.$$

Thus $a^d \in K^n$, as $\mathrm{norm}_{K(\alpha)/K}(\alpha) \in K$. If $a^k \in K^n$, say $a^k = c^n$ for some $c \in K$, then

$$c^n = a^k = (\alpha^n)^k = (\alpha^k)^n \implies \alpha^k = c\omega \in K$$

for some $\omega \in G_n(K)$. Thus α is a root of $X^k - \alpha^k \in K[X]$ and hence $k \geq d$.

Note that $f(\alpha, K) = X^d - \alpha^d$.

Let E/K be cyclic of degree n. Assume that $\mathrm{Gal}(E/K) = \langle\sigma\rangle$. If ω is a primitive n-root of one,

$$\mathrm{norm}_{E/K}(\omega) = \omega^n = 1.$$

By Hilbert's theorem 90, there exists a $b \in E^\times$ such that $\omega = b/\sigma(b)$. Thus $\sigma(b) = \omega^{-1}b$ and hence $\sigma^i(b) = \omega^{-i}b$ for all $i \geq 0$. Since $|\{b, \sigma(b), \dots, \sigma^{n-1}(b)\}| = n$, it follows that $E = K(b)$. Moreover,

$$\sigma(b^n) = \sigma(b)^n = (\omega^{-1}b)^n = b^n$$

and hence $b^n \in K$. This means that E/K is a decomposition field of $X^n - b^n$. Note that $X^n - b^n$ is irreducible, as $[E : K] = [K(b) : K] = n$. □

16.6. Proposition. *Let K be a field of characteristic $p > 0$.*

1) *Let $a \in K$ and $f = X^p - X - a$. Then f is irreducible over K or all the roots of f belong to K. In the first case, if b is a root of f, then $K(b)/K$ is a cyclic extension of degree p.*
2) *Every cyclic extension of degree p is a splitting field of an irreducible polynomial of the form $X^p - X - a$.*

Proof. We first prove 1). Let K_0 be the prime field of K and C be an algebraic closure of K. Note that $K_0 \simeq \mathbb{Z}/p$. Let $b \in C$ be a root of f and let $x \in K_0$. Then

$$f(b+x) = (b+x)^p - (b+x) - a = (b^p - b - a) + (x^p - x) = 0$$

and thus $\{b + x : x \in K_0\} \subset C$ is the set of roots of f. Note that $f' = -1$, so f has no multiple roots.

We claim that if $b \notin K$, then f is irreducible. If f is not irreducible, then $f = gh$ for some $g, h \in K[X]$ such that $0 < \deg g < p$. There exists a subset S of K_0 such that $g = \prod_{x \in S}(X - (b + x))$ in $C[X]$ and hence

$$|S|b + \sum_{x \in S} x = \sum_{x \in S}(b + x) \in K.$$

This implies that $|S|b \in K$ and hence, since $|S| \in K^\times$, it follows that $b \in K$.

Since $K(b)/K$ is a splitting field of a separable polynomial, $K(b)/K$ is a Galois extension. Moreover, $|\operatorname{Gal}(K(b)/K)| = [K(b) : K] = p$ and hence $\operatorname{Gal}(K(b)/K)$ is cyclic.

We now prove 2). Let E/K be cyclic of degree p. Assume that $\operatorname{Gal}(E/K) = \langle \sigma \rangle$. Since $\operatorname{trace}_{E/K}(1) = p = 0$, Hilbert's theorem implies that there exists $b \in E$ such that $\sigma(b) = b+1$. In particular, $b \notin K$ and thus $E = K(b)$. Moreover, since

$$\sigma(b^p - b) = \sigma(b)^p - \sigma(b) = (b + 1)^p - (b + 1) = b^p - b,$$

it follows that $b^p - b \in K$. Thus $f(b, K) = X^p - X - (b^p - b) \in K[X]$. $\square$

§ 17. Symmetric polynomials

Let K be a field and $\{t_1, \dots, t_n\}$ be a commuting set of independent variables. Let $E = K(t_1, \dots, t_n)$ and $f = \prod_{i=1}^n (X - t_i) \in E[X]$. Then

$$f = X^n + \sum_{i=1}^n (-1)^i s_i X^{n-i},$$

where

$$\begin{aligned} s_1 &= t_1 + t_2 + \cdots + t_n, \\ s_2 &= \sum_{1 \leq i < j \leq n} t_i t_j, \\ &\vdots \\ s_n &= t_1 t_2 \cdots t_n. \end{aligned}$$

For example,

$$(X - t_1)(X - t_2)(X - t_3) = X^3 - (t_1 + t_2 + t_3)X^2 + (t_1t_2 + t_2t_3 + t_1t_3)X - t_1t_2t_3.$$

The polynomials $s_1, s_2, \dots, s_n$ are known as the **elementary symmetric polynomials** in the variables $t_1, \dots, t_n$. Note that $\deg s_i = i$.

Let $\sigma \in \mathbb{S}_n$ and

$$\alpha_\sigma \colon K[t_1, \dots, t_n] \to K[t_1, \dots, t_n], \quad t_i \mapsto t_{\sigma(i)} \quad \text{for all } i.$$

Then α_σ is a bijective homomorphism of K-algebras. In fact, $\alpha_\sigma^{-1} = \alpha_{\sigma^{-1}}$. Note that

$$\alpha_\sigma(h(t_1, \dots, t_n)) = h(t_{\sigma(1)}, \dots, t_{\sigma(n)}).$$

Since α_σ is injective, it induces an element $\widehat{\sigma} \in \operatorname{Gal}(E/K)$ given by

$$\widehat{\sigma}\left(\frac{h}{g}\right) = \frac{\alpha_\sigma(h)}{\alpha_\sigma(g)}.$$

The map $\mathbb{S}_n \to \operatorname{Gal}(E/K)$, $\sigma \mapsto \widehat{\sigma}$, is an injective group homomorphism. Thus

$$\{\widehat{\sigma} : \sigma \in \mathbb{S}_n\} \simeq \mathbb{S}_n.$$

§ 17. Symmetric polynomials

17.1. Definition. Let $g \in K[t_1, \dots, t_n]$. Then g is **symmetric** if $\widehat{\sigma}(g) = g$ for all $\sigma \in \mathbb{S}_n$.

We write P to denote the set of symmetric polynomials in $K[t_1, \dots, t_n]$. Clearly, P is a subalgebra of $K[t_1, \dots, t_n]$. The following statements hold:

1) $K \subseteq P$.
2) $\sum_{i=1}^{n} t_i^r \in P$ for all $r \geq 1$.
3) $s_i \in P$ for all i.
4) $K(P) \subseteq {}^G E$, where $G = \{\widehat{\sigma} : \sigma \in \mathbb{S}_n\}$.

Let $F = K(s_1, s_2, \dots, s_n)$. Then E/F is a Galois extension, as it is a splitting field of f.

17.2. Proposition. $[E : F] \leq n!$.

Proof. We proceed by induction on n. The case $n = 1$ is clear, as $E = F$. Assume that $n > 1$. Let $u_1, \dots, u_{n-1}$ be the elementary symmetric polynomials in $t_1, \dots, t_{n-1}$. Then

$$s_i = u_i + t_n u_{i-1}$$

for all $i \in \{1, \dots, n\}$, where $u_0 = 1$ and $u_n = 0$. Note that $u_1 = s_1 - t_n$ and $u_i = s_i - t_n u_{i-1}$ for all i. Since $K(s_1, \dots, s_n, t_n) = K(u_1, \dots, u_{n-1}, t_n)$,

$$F(t_n) = K(u_1, \dots, u_{n-1}, t_n) = K(t_n)(u_1, \dots, u_{n-1})$$

and

$$[E : F] = [E : F(t_n)][F(t_n) : F] \leq n[E : F(t_n)].$$

Note that $E = K(t_1, \dots, t_n) = K(t_n)(t_1, \dots, t_{n-1})$. By the inductive hypothesis,

$$[E : F(t_n)] \leq (n-1)!$$

and hence $[E : F] \leq n!$. $\square$

17.3. Theorem. ${}^G E = F$.

Proof. By Artin's theorem,

$$\left[{}^G E : F\right] = \frac{[E : F]}{\left[E : {}^G E\right]} \leq \frac{n!}{\left[E : {}^G E\right]} = 1$$

and hence ${}^G E = F$. $\square$

17.4. Exercise. Prove that $\operatorname{Gal}(E/F) \simeq \mathbb{S}_n$.

17.5. Exercise. Prove that $\{s_1, \dots, s_n\}$ is algebraically independent over K.

17.6. Exercise. Prove that every symmetric polynomial in $t_1, \dots, t_n$ can be written as a rational fraction in $s_1, \dots, s_n$.

§ 18. Solvable groups

Let G be a group. If $x, y \in G$ we define the **commutator** of x and y as

$$[x, y] = xyx^{-1}y^{-1}.$$

Note that $[x, y] = 1$ if and only if $xy = yx$. Moreover, $[x, y]^{-1} = [y, x]$. The **commutator (or derived) subgroup** $[G, G]$ of G is defined as the subgroup of G generated by all commutators, i.e.

$$[G, G] = \langle [x, y] : x, y \in G \rangle.$$

This means that every element of $[G, G]$ is a finite product of commutators, so every element of $[G, G]$ is of the form $\prod_{i=1}^{m}[x_i, y_i]$. In general, the commutator subgroup is not equal to the set of commutators!

18.1. Example. This example is taken from the book [**2**] of Carmichael. Let G be the subgroup of $\mathbb{S}_{16}$ generated by the permutations

$$\begin{aligned} a &= (13)(24), & b &= (57)(68), \\ c &= (911)(10\,12), & d &= (13\,15)(14\,16), \\ e &= (13)(57)(9\,11), & f &= (12)(34)(13\,15), \\ g &= (56)(78)(13\,14)(15\,16), & h &= (9\,10)(11\,12). \end{aligned}$$

Then $[G, G]$ has order 16. However, the set $\{[x, y] : x, y \in G\}$ of commutators has 15 elements:

```
> S16 := Sym(16);
> a := S16 ! (1,3)(2,4);
> b := S16 ! (5,7)(6,8);
> c := S16 ! (9,11)(10,12);
> d := S16 ! (13,15)(14,16);
> e := S16 ! (1,3)(5,7)(9,11);
> f := S16 ! (1,2)(3,4)(13,15);
> g := S16 ! (5,6)(7,8)(13,14)(15,16);
> h := S16 ! (9,10)(11,12);
> G := PermutationGroup< 16 | a,b,c,d,e,f,g,h >;
> D := DerivedSubgroup(G);
> #D;
16
> #{ x*y*Inverse(x)*Inverse(y) : x in G, y in G };
15
> c*d in { x : x in D } \
> diff { u*v*Inverse(u)*Inverse(v) : u in G, v in G };
true
```

18.2. Exercise. Let G be a group. Prove the following facts:

1) G is abelian if and only if $[G,G]=\{1\}$.
2) $[G,G]$ is a normal subgroup of G.
3) $G/[G,G]$ is abelian.
4) If H is a subgroup of G and $[G,G]\subseteq H$, then H is normal in G.
5) If H is a normal subgroup of G, then G/H is abelian if and only if $[G,G]\subseteq H$.

18.3. Definition. Let G be a group. The **derived series** of G is defined as $G^{(0)}=G$ and $G^{(k+1)}=[G^{(k)},G^{(k)}]$ for $k\geq 0$.

18.4. Exercise. Prove that $G^{(k)}$ is normal in G for all k.

Why derived series? We cannot explain this here, but let us use the following notation. We write $G'=[G,G]$, $G''=[G',G']$... Note that

$$G\supseteq G'\supseteq G''\supseteq\cdots$$

18.5. Exercise. Let $n\geq 3$. Prove that $[\mathbb{S}_n,\mathbb{S}_n]=\mathbb{A}_n$.

18.6. Example. Let $K=\{\mathrm{id},(12)(34),(13)(24),(14)(23)\}$. Then K is a normal subgroup of $\mathbb{A}_4$. One proves that $[\mathbb{A}_4,\mathbb{A}_4]=K$.

A group G is said to be **simple** if there are no proper non-trivial normal subgroups of G. If p is a prime number, then the group $\mathbb{Z}/p$ of integers modulo p is a simple group. We will prove later that $\mathbb{A}_n$ is simple if $n\geq 5$.

18.7. Example. Let $n\geq 5$. Since $\mathbb{A}_n$ is a non-abelian simple group, $[\mathbb{A}_n,\mathbb{A}_n]=\mathbb{A}_n$.

Let us show that $\mathbb{A}_5$ is a non-abelian simple group. Hence it is not solvable:

```
> G := Alt(5);
> IsAbelian(G);
false
> IsSimple(G);
true
> IsSolvable(G);
false
```

18.8. Definition. A group G is **solvable** if and only if $G^{(m)}=\{1\}$ for some m.

Every abelian group is solvable.

18.9. Exercise. Prove that $\mathbb{S}_n$ is solvable if and only if $n\leq 4$.

Let us compute (with the computer software Oscar) the derived series of the symmetric group $\mathbb{S}_4$. The calculation shows that $\mathbb{S}_4$ is solvable:

```
> G := Sym(4);
> DerivedSeries(G);
[
    Symmetric group G acting on a set of cardinality 4
```

```
    Order = 24 = 2^3 * 3
        (1, 2, 3, 4)
        (1, 2),
    Permutation group acting on a set of cardinality 4
    Order = 12 = 2^2 * 3
        (1, 2, 3)
        (2, 3, 4),
    Permutation group acting on a set of cardinality 4
    Order = 4 = 2^2
        (1, 4)(2, 3)
        (1, 3)(2, 4),
    Permutation group acting on a set of cardinality 4
    Order = 1
]
> IsSolvable(G);
true
```

18.10. Proposition. *Let G be a group and H be a subgroup of G. The following statements hold:*

1) *If G is solvable, then H is solvable.*
2) *If H is normal in G and G is solvable, then G/H is solvable.*
3) *If H is normal in G and H and G/H are solvable, then G is solvable.*

Proof. The first statement follows from the fact that $H^{(i)} \subseteq G^{(i)}$ holds for all i.

Assume now that H is normal in G. Let $Q = G/H$ and $\pi\colon G \to Q$ be the canonical map. By induction one proves that $\pi(G^{(i)}) = Q^{(i)}$ for all $i \geq 0$. The case where $i = 0$ is trivial, as π is surjective. If the result holds for some $i \geq 0$, then

$$\pi(G^{(i+1)}) = \pi([G^{(i)}, G^{(i)}]) = [\pi(G^{(i)}), \pi(G^{(i)})] = [Q^{(i)}, Q^{(i)}] = Q^{(i+1)}.$$

We now prove 2). Since G is solvable, $G^{(n)} = \{1\}$ for some n. Thus Q is solvable, as $Q^n = \pi(G^{(n)}) = \pi(\{1\}) = \{1\}$.

We finally prove 3). Since Q is solvable, $Q^{(n)} = \{1\}$ for some n. Moreover, since $\pi(G^{(n)}) = Q^{(n)} = \{1\}$, it follows that $G^{(n)} \subseteq H$. Since H is solvable,

$$G^{(n+m)} \subseteq (G^{(n)})^{(m)} \subseteq H^{(m)} = \{1\}$$

for some m. Thus G is solvable. □

An application:

18.11. Proposition. *Let G be a finite p-group. Then G is solvable.*

Proof. Assume the result is not true. Let G be a finite p-group of minimal order that is not solvable. Since G is a p-group, $Z(G) \neq \{1\}$. Since $|G|$ is minimal, $G/Z(G)$ is a solvable p-group. Since $Z(G)$ is abelian, $Z(G)$ is solvable. Now G is solvable by Proposition 18.10. □

Let G be a group. A proper subgroup N of G is said to be **maximal normal** if N is a normal subgroup of G and there is no other proper normal subgroup of G containing N.

18.12. EXERCISE. If a subgroup N of G is maximal (for the inclusion) and normal, then it is maximal normal. Show that the converse does not hold.

The following result is a direct consequence of the correspondence theorem:

18.13. EXERCISE. Let G be a group and N be a normal subgroup of G. Prove that N is maximal normal if and only if G/N is simple.

Maximal normal subgroups always exist in finite groups (they could be trivial). We can compute maximal normal subgroups as follows:

```
> MaximalNormalSubgroup(Sym(3));
Permutation group acting on a set of cardinality 3
Order = 3
    (1, 2, 3)
julia> maximal_normal_subgroups(quaternion_group(8))
3-element Vector{PcGroup}:
 Group([ y2, x ])
 Group([ y2, y ])
 Group([ y2, x*y ])

> MaximalNormalSubgroup(Alt(4));
Permutation group acting on a set of cardinality 4
Order = 4 = 2^2
    (1, 2)(3, 4)
    (1, 3)(2, 4)
```

18.14. EXERCISE. Let G be a finite solvable group. Prove that if G is simple, then G is cyclic of prime order.

The following result will be important later:

18.15. PROPOSITION. *Every finite solvable group contains a normal subgroup of prime index.*

PROOF. Let G be a finite solvable group. Let M be a maximal normal subgroup of G (there is at least one, as G is finite). Since G/M is simple and solvable (see Proposition 18.10), G/M is cyclic of prime order by Exercise 18.14. $\square$

We finish this discussion with two important theorems (without proof) about finite solvable groups.

18.16. THEOREM (Burnside). *Let p and q be prime numbers. If G is a group of order $p^a q^b$, then G is solvable.*

The proof appears in courses on the representation theory of finite groups.

18.17. THEOREM (Feit–Thompson). *Every finite group of odd order is solvable.*

The proof of the theorem is extremely hard. It occupies a full volume of **Pacific Journal of Mathematics**, see [**3**].

The proof is hard and occupies a full volume [**3**] of the **Pacific Journal of Mathematics** (see Figure 18.1). The theorem was formally verified by the computer proof assistant `Coq`, see [**?**] for the announcement.

The proof of the Feit–Thompson theorem suggests the following conjecture:

18.18. Open problem (Feit–Thompson). Are there distinct prime numbers p and q such that $\frac{p^q-1}{p-1}$ divides $\frac{q^p-1}{q-1}$?

The conjecture states that there are no such prime numbers. The problem is still open; see [**?**] for some partial results.

Figure 18.1. The proof of the Feit–Thompson theorem occupies a full volume of the Pacific Journal of Mathematics.

The Feit–Thompson theorem is one of the starting points of the classification of finite simple groups (CFSG). This classification is one of the deepest theorems of the 20th century. The proof consists of tens of thousands of pages in several hundred journal articles written by about 100 authors, published mostly between 1955 and 2004.

18.19. Theorem (CFSG). *Let G be a finite simple group. Then G lies in one (or more) of the following families:*

1) *Cyclic groups of prime order.*
2) $\mathbb{A}_n$ *for* $n \geq 5$.
3) *Finite groups of Lie type.*
4) *26 sporadic simple groups.*

Groups of Lie type are finite analogs of simple Lie groups, such as $\mathbf{SL}_n(\mathbb{C})$. A typical example of a finite simple group of Lie type is $\mathbf{PSL}_n(p)$ for some prime number p, which is defined as $\mathbf{SL}_n(p)$ over its center.

The **sporadic simple groups** are 26 groups that do not follow a systematic pattern. The first five of the sporadic groups were discovered by Mathieu in the 1860s, and the other 21 were found between 1965 and 1975. Several of these groups were predicted to exist before they were constructed, sometimes just knowing their character tables! The full list of sporadic simple groups is as follows:

- Mathieu groups M_{11}, M_{12}, M_{22}, M_{23} and M_{24}.

- Janko groups J_1, J_2, J_3 and J_4.
- Conway groups Co_1, Co_2 and Co_3.
- Fischer groups Fi_{22}, Fi_{23} and Fi_{24}.
- Higman–Sims group HS.
- McLaughlin group McL.
- Held group He.
- Rudvalis group Ru.
- Suzuki group Suz.
- O'Nan group ON.
- Harada–Norton group HN.
- Lyons group Ly.
- Thompson group Th.
- Baby Monster group B.
- Monster group M.

Mathieu groups can be realized as automorphism groups of certain very complicated combinatorial structures known as **Steiner systems**. A concrete example:

$$M_{11} = \langle (123456789\,10\,11), (37\,11\,8)(4\,10\,56) \rangle,$$

which is a subgroup of $\mathbb{S}_{11}$ of order 7920.

§ 19. Simplicity of the alternating simple group

We will present a family of non-abelian simple groups. We start with some exercises.

19.1. Exercise. Let G be a group. Prove that G is simple if and only if $\{(g,g) : g \in G\}$ is a maximal subgroup of $G \times G$.

19.2. Exercise. Prove that $\mathbb{A}_n$ is generated by 3-cycles.

19.3. Exercise. Compute the commutator subgroup of $\mathbb{A}_n$ for $n \geq 2$.

Note that $\mathbb{A}_2$ and $\mathbb{A}_3$ are abelian. For $\mathbb{A}_4$, one proves that

$$[\mathbb{A}_4, \mathbb{A}_4] = \{\mathrm{id}, (12)(34), (13)(24), (14)(23)\}.$$

Finally, $[\mathbb{A}_n, \mathbb{A}_n] = \mathbb{A}_n$ for $n \geq 5$.

Let us compute some commutator subgroups (and the inclusion group homomorphism) with the computer:

```
> S3 := Sym(3);
> DerivedSubgroup(S3);
Permutation group acting on a set of cardinality 3
Order = 3
    (1, 2, 3)
```

19.4. Exercise. Let $n \geq 3$. Prove that $[\mathbb{S}_n, \mathbb{S}_n] = \mathbb{A}_n$.

§ 19. Simplicity of the alternating simple group

Recall that every normal subgroup is a union of conjugacy classes. The group $\mathbb{A}_5$ has conjugacy classes of sizes 1, 15, 20, 12 and 12. It follows that the only possible normal subgroups of $\mathbb{A}_5$ are $\{\mathrm{id}\}$ and $\mathbb{A}_5$.

```
> A5 := Alt(5);
> ConjugacyClasses(A5);
Conjugacy Classes of group A5
-----------------------------
[1]     Order 1       Length 1
        Rep Id(A5)

[2]     Order 2       Length 15
        Rep (1, 2)(3, 4)

[3]     Order 3       Length 20
        Rep (1, 2, 3)

[4]     Order 5       Length 12
        Rep (1, 2, 3, 4, 5)

[5]     Order 5       Length 12
        Rep (1, 3, 4, 5, 2)
```

19.5. Theorem (Jordan). *Let $n \geq 5$. Then $\mathbb{A}_n$ is simple.*

Before proving the theorem, we need some preliminary results.

Every permutation $\rho \in \mathbb{S}_n$ decomposes as a product of disjoint cycles, say

$$\rho = (a_1 \cdots a_r)(b_1 \cdots b_s) \cdots (c_1 \cdots c_t)$$

where, by convention, we do not write cycles of length one. The cyclic structure of ρ is, by definition, the (unordered) sequence of integers $r, s, \dots t$, where, again by convention, we omit fixed points. For example, the cyclic structure of the transposition (ab) is 2, of $(abc)(d)$ is 3 and of $(123)(45)(789a)(bcd)(d)$ is 2,3,3,4.

19.6. Lemma. *If both permutations $\rho_1 \in \mathbb{S}_n$ and $\rho_2 \in \mathbb{S}_n$ have the same cyclic structure, then $\rho_2 = \sigma\rho_1\sigma^{-1}$ for some $\sigma \in \mathbb{S}_n$.*

Proof. Assume that

$$\rho_1 = (a_1 \cdots a_r)(b_1 \cdots b_s) \cdots (c_1 \cdots c_t),$$
$$\rho_2 = (x_1 \cdots x_r)(y_1 \cdots y_s) \cdots (z_1 \cdots z_t).$$

Let

$$\mathrm{Fix}(\rho_1) = \{x \in \{1, \dots, n\} : \rho_1(x) = x\} = \{k_1, \dots, k_m\}, \qquad \mathrm{Fix}(\rho_2) = \{l_1, \dots, l_m\}$$

be the fixed points of the permutations ρ_1 and ρ_2, respectively. Then

$$\sigma(x) = \begin{cases} x_j & \text{if } x = a_j \text{ for some } j, \\ y_j & \text{if } x = b_j \text{ for some } j, \\ \vdots & \\ z_j & \text{if } x = c_j \text{ for some } j, \\ l_j & \text{if } x = k_j \text{ for some } j, \end{cases}$$

is such that $\sigma\rho_1\sigma^{-1} = \rho_2$. □

What happens with the alternating group?

19.7. Lemma. *If $\rho_1, \rho_2 \in \mathbb{S}_n$ are conjugate in $\mathbb{S}_n$ and $|\operatorname{Fix}(\rho_1)| \geq 2$, then $\mu\rho_1\mu^{-1} = \rho_2$ for some $\mu \in \mathbb{A}_n$.*

Proof. Assume that $\rho_2 = \sigma\rho_1\sigma^{-1}$ for some $\sigma \in \mathbb{S}_n$. There are $a, b \in \{1, \dots, n\}$ such that $\rho_1(a) = a$, $\rho_1(b) = b$ and $a \neq b$. Let

$$\mu = \begin{cases} \sigma & \text{if } \sigma \in \mathbb{A}_n, \\ \sigma(ab) & \text{otherwise.} \end{cases}$$

Then $\mu \in \mathbb{A}_n$ and $\mu\rho_1\mu^{-1} = \rho_2$, as (ab) commutes with ρ_1. □

Let us discuss some examples.

19.8. Example. If $\rho_1 = (23)(156)$ and $\rho_2 = (45)(123)$, then $\rho_2 = \sigma\rho_1\sigma^{-1}$ for

$$\sigma = \begin{pmatrix} 123456 \\ 145623 \end{pmatrix}.$$

19.9. Example. The permutations $\rho_1 = (123) \in \mathbb{S}_3$ and $\rho_2 = (132) \in \mathbb{S}_3$ are conjugate in $\mathbb{S}_3$, as $(123) = \sigma(132)\sigma^{-1}$ if $\sigma = (23)$. However, ρ_1 and ρ_2 are not conjugate in $\mathbb{A}_3$.

Now we are ready to prove the theorem.

Proof of Theorem 19.5. Let $N \neq \{\mathrm{id}\}$ be a normal subgroup of $\mathbb{A}_n$. If $(abc) \in N$, then every 3-cycle belongs to N, because all 3-cycles are conjugate in $\mathbb{S}_n$, and the previous lemma states that $(ijk) = \mu(abc)\mu^{-1} \in N$ for some $\mu \in \mathbb{A}_n$. Thus $N = \mathbb{A}_n$.

We claim that N contains a 3-cycle. Since $N \neq \{\mathrm{id}\}$, there exists $\sigma \in N \setminus \{\mathrm{id}\}$. Let $m = |\sigma|$ and let p be a prime number dividing m. Then $\tau = \sigma^{m/p}$ has order p and hence $\tau = \rho_1 \cdots \rho_s$, where the ρ_j's are disjoint p-cycles.

If $p = 2$, then $1 = \operatorname{sign}(\tau) = (-1)^s$. Thus s is even. Write

$$\tau = (ab)(cd)\rho_3 \cdots \rho_s.$$

Since $\rho_3 \cdots \rho_s$ commutes with (abc) and (acb),

$$\underbrace{(abc)\tau(abc)^{-1}\tau^{-1}}_{\in N} = (abc)(ab)(cd)(acb)(ab)(cd) = (ac)(bd).$$

Hence $(ac)(bd) \in N$. Let $e \in \{1, \dots, n\} \setminus \{a, b, c, d\}$. Then

$$(ae)(bd) = (aec)\underbrace{(ac)(bd)}_{\in N}(aec)^{-1} \in N$$

and therefore
$$(aec) = (ac)(ae) = (ac)(bd)(ae)(bd) \in N.$$

If $p = 3$, without loss of generality, we may assume that $s \geq 2$ (otherwise, τ would be a 3-cycle). Then $\tau = (abc)(def)\rho_3 \cdots \rho_s$. Since (bcd) commutes with $\rho_3 \cdots \rho_s$ and N is normal in $\mathbb{A}_n$,
$$\underbrace{(bcd)\tau(bcd)^{-1}\tau^{-1}}_{\in N} = (bcd)(abc)(def)(bdc)(acb)(dfe) = (adbce)$$
and therefore
$$(adc) = (adb)(adbce)(adb)^{-1}(adbce)^{-1} \in N.$$

If $p > 3$, then $\tau = (abcd \cdots z)\rho_2 \cdots \rho_s$. In particular, (abc) commutes with $\rho_2 \cdots \rho_s$. Then
$$(abd) = (abc)\tau(abc)^{-1}\tau^{-1} \in N.$$ □

As an application, we compute the normal subgroups of the symmetric group $\mathbb{S}_n$.

19.10. Exercise. Compute the list of normal subgroups of $\mathbb{S}_n$ for $n \geq 2$.

§ 20. Radical extensions

20.1. Definition. An extension E/K is said to be **pure** of type m if $E = K(x)$ for some x such that $x^m \in K$.

Note that if $E = K(x)$ is a pure extension of type m and K contains all m-th roots of one, then E/K is a splitting field of $X^m - x^m$.

20.2. Definition. The sequence $K = R_0 \subseteq R_1 \subseteq \cdots \subseteq R_m$ of fields is said to be a **radical tower** if each R_{i+1}/R_i is pure. In this case, R_m/K is a **radical extension**.

Note that radical extensions are finite.

20.3. Example. Let E be a splitting field of $X^4 - 2$ over $\mathbb{Q}$. Then $E/\mathbb{Q}$ is radical, as $E = \mathbb{Q}(\sqrt[4]{2}, i)$.

20.4. Example. Let $\alpha, \beta \in \mathbb{C}$ be such that $\alpha^2 = 2$ and $\beta^5 = 1+\alpha$. The number $\sqrt[5]{1+\sqrt{2}}$ belongs to the radical extension $\mathbb{Q}(\alpha, \beta)/\mathbb{Q}$.

20.5. Theorem. *Let K be of characteristic zero and R/K be a radical extension. If E/K is a subextension of R/K, then* $\operatorname{Gal}(E/K)$ *is solvable.*

Proof. Without loss of generality, we may assume that E/K is a Galois extension. To prove this fact, let $G = \operatorname{Gal}(E/K)$ and $F = {}^{G}E$. Then E/F is a Galois extension and $\operatorname{Gal}(E/F) = G$ by Artin's theorem. Thus, replacing K by F if needed, we may assume that E/K is Galois.

Let L be the normal closure of R in some algebraic closure C that contains R. Note that if $R = K(x_1, \ldots, x_m)$, then
$$L = K(\{\sigma_i(x_j) : 1 \leq i \leq s,\ 1 \leq j \leq m\}),$$
where $\operatorname{Hom}(R/K, C/K) = \{\sigma_1, \ldots, \sigma_s\}$.

Claim. L/K is radical.

Since $x_j^{a_j} \in K(x_1, \dots, x_{j-1})$ for some integer a_j,

$$\sigma_i(x_j)^{a_j} = \sigma_i\left(x_j^{a_j}\right) \in \sigma_i(K(x_1, \dots, x_{j-1})) = K(\sigma_i(x_1), \dots, \sigma_i(x_{j-1}))$$

Thus L/K is radical and Galois.

We may assume then that E/K and R/K are both Galois.

Since $\mathrm{Gal}(E/K) \simeq \mathrm{Gal}(R/K)/\mathrm{Gal}(R/E)$, we only need to prove that $\mathrm{Gal}(R/K)$ is solvable.

For a positive integer n, let ξ be a primitive n-th root of one (in some algebraic closure of K that contains R). Consider the diagram

$$\begin{tikzcd}[column sep=small]
& R(\xi) \arrow[dl, dash] \arrow[dr, dash] & \\
R \arrow[dr, dash] & & K(\xi) \arrow[dl, dash] \\
& K &
\end{tikzcd}$$

Then

1) $K(\xi)/K$ and $R(\xi)/R$ are abelian.
2) $R(\xi)/K$ is Galois.
3) $\mathrm{Gal}(R/K) \simeq \mathrm{Gal}(R(\xi)/K)/\mathrm{Gal}(R(\xi)/R)$.
4) $\mathrm{Gal}(K(\xi)/K) \simeq \mathrm{Gal}(R(\xi)/K)/\mathrm{Gal}(R(\xi)/K(\xi))$.

The third item implies that we need to show that $\mathrm{Gal}(R(\xi)/K)$ is solvable. By the fourth item, it suffices to show that $\mathrm{Gal}(R(\xi)/K(\xi))$ is solvable (because $\mathrm{Gal}(K(\xi)/K)$ is abelian and hence solvable).

Since $R = K(x_1, \dots, x_m)$,

$$R(\xi) = K(x_1, \dots, x_m, \xi) = K(\xi)(x_1, \dots, x_m)$$

and hence $R(\xi)/K(\xi)$ is radical. This means that without loss of generality, we may assume that K contains primitive n-roots of one. For example, if $R = K(x_1, \dots, x_m)$ and $x_i^{a_i} \in K(x_1, \dots, x_{i-1})$, then we may assume that K contains a primitive a_i-root of one. We proceed by induction on m. The case $m = 0$ is trivial. Assume that the claim holds for some $m \geq 0$. Let $L = K(x_1)$. Then L/K is a splitting field of $X^{a_1} - x_1^{a_1}$, and hence L/K is a cyclic extension. Thus $\mathrm{Gal}(L/K)$ is cyclic (and hence, in particular, solvable). Let H be the subgroup that corresponds to L, that is $H = \mathrm{Gal}(R/L)$ (here, we use Galois' correspondence). Then H is normal in $\mathrm{Gal}(R/K)$. Since $R = K(x_1, \dots, x_m) = L(x_2, \dots, x_m)$, R/L is radical and Galois. By the inductive hypothesis, $\mathrm{Gal}(R/L)$ is solvable. Since

$$\mathrm{Gal}(L/K) \simeq \mathrm{Gal}(R/K)/\mathrm{Gal}(R/L),$$

it follows that $\mathrm{Gal}(R/K)$ is solvable. $\square$

20.6. Definition. Let $f \in K[X]$ and E be a splitting field of f over K. We say that f is **solvable by radicals** if there is a radical extension R/K such that $E \subseteq R$.

The general polynomial of degree two is solvable by radicals, as its Galois group is solvable (in fact, isomorphic to $\mathbb{S}_2$).

20.7. Exercise. Prove that $f = X^2 - s_1X + s_2 \in \mathbb{Q}[X]$ is solvable by radicals.

Theorem 20.5 translates into the following result:

20.8. Exercise. Let K be a field of characteristic zero. If $f \in K[X]$ is solvable by radicals, then $\mathrm{Gal}(f, K)$ is solvable.

As a consequence, the general polynomial of degree $n \geq 5$ is not solvable by radicals, as its Galois group contains a subgroup isomorphic to $\mathbb{S}_5$.

20.9. Example. Let p be a prime number and $f = X^5 - 2pX + p \in \mathbb{Q}[X]$. We claim that f is not solvable by radicals.

Note that $f' = 5X^4 - 2p$. Then $\alpha = \sqrt[4]{2p/5}$ and $\beta = -\sqrt[4]{2p/5}$ are critical points. Since $f(\alpha) < 0$ and $f(\beta) > 0$, it follows that f has exactly three real roots. Let $x_1, x_2 \in \mathbb{C} \setminus \mathbb{R}$ and $x_3, x_4, x_5 \in \mathbb{R}$ be the roots of f.

By Eisenstein's theorem, f is irreducible.

Let $E/\mathbb{Q}$ be a splitting field of f. Then $\mathrm{Gal}(f, \mathbb{Q}) = \mathrm{Gal}(E/\mathbb{Q})$ is isomorphic to a subgroup G of $\mathbb{S}_5$. Since f is irreducible, 5 divides $[E : \mathbb{Q}] = |G|$. In particular, by Cauchy's theorem, G contains an element σ of order five. This element is a 5-cycle, so without loss of generality, we may assume that $\sigma = (x_1x_2x_3x_4x_5)$. Note that $(x_1x_2) \in G$. Thus $G \simeq \mathbb{S}_5$ and hence G is not solvable.

20.10. Exercise. Let $f = X^6 + 2X^5 - 5X^4 + 9X^3 - 5X^2 + 2X + 1 \in \mathbb{Q}[X]$. Prove that f is solvable by radicals.

It is now time to prove Galois' great theorem on solvability of polynomials.

20.11. Theorem (Galois). *Let K be a field of characteristic zero and $f \in K[X]$. Then f is solvable by radicals if and only if* $\mathrm{Gal}(f, K)$ *is solvable.*

We proved in Theorem 20.5 that solvable polynomials have solvable Galois groups. For the converse, we need two auxiliary results.

20.12. Lemma. *Let E/K be a Galois extension of prime degree p. Assume that K admits a primitive p-root of one. Then $E = K(\beta)$ where $\beta^p \in K$.*

Proof. Assume that $\mathrm{Gal}(E/K) = \langle\sigma\rangle$. Let $\omega \in K$ be a primitive p-root of one. Then $\mathrm{norm}_{E/K}(\omega) = \omega^p = 1$. By Hilbert's theorem, $\omega = \beta/\sigma(\beta)$ for some $\beta \in E$. Note that $\beta \notin K$, as $\omega \neq 1$. Moreover,

$$\sigma(\beta^p) = (\beta\omega^{-1})^p = \beta^p \in {}^{\mathrm{Gal}(E/K)}E = K.$$

Since $K \subseteq K(\beta) \subseteq E$ and $[E : K] = p$, we conclude that $E = K(\beta)$ with $\beta^p \in K$. □

20.13. Exercise. Let E/K be a splitting field of $f \in K[X]$ and K^*/K be an extension. If E^*/K^* is a splitting field of f containing E, then

$$\mathrm{Gal}(E^*/K^*) \to \mathrm{Gal}(E/K), \quad \sigma \mapsto \sigma|_E,$$

is an injective group homomorphism.

Now Theorem 20.11 will follow from the following theorem.

20.14. Theorem. *Let K be a field of characteristic zero and E/K be a finite Galois extension. If* $\mathrm{Gal}(E/K)$ *is solvable, then E can be embedded in a radical extension.*

Proof. Let $G = \mathrm{Gal}(E/K)$. Since G is solvable, by Proposition 18.15, there exists a normal subgroup H of G of prime index p. Let ω be a primitive p-th root of one. (It exists because K is a field of characteristic zero.)

We proceed by induction on $[E : K]$.

If $[E : K] = 1$, there is nothing to prove. So assume that $[E : K] > 1$.

We first assume that $\omega \in K$. The group $\mathrm{Gal}(E/{}^HE)$ is solvable, as it is a subgroup of G. Moreover, since

$$[E : {}^HE] < [E : K],$$

the inductive hypothesis implies that $E/{}^HE$ can be embedded in a radical extension, so there exists a radical tower

$$ {}^HE \subseteq R_1 \subseteq R_2 \subseteq \cdots \subseteq R_m, \tag{20.1}$$

where $E \subseteq R_m$. Now ${}^HE/K$ is a Galois extension, as H is normal in G. Moreover,

$$[{}^HE : K] = (G : H) = p.$$

Since $\omega \in K$, Lemma 20.12 implies that ${}^HE = K(\beta)$ for some β such that $\beta^p \in K$. The radical tower 20.1 can be extended by adding $K \subseteq {}^HE$.

For the general case, let $K^* = K(\omega)$ and $E^* = E(\omega)$. Then E^*/K^* is a Galois extension with Galois group $\mathrm{Gal}(E^*/K^*)$. By Exercise 20.13, $\mathrm{Gal}(E^*/K^*)$ is solvable. By the previous part, E^* and E can be embedded in a radical extension R^*/K^*, so there exists a radical tower

$$K^* \subseteq R_1^* \subseteq R_2^* \subseteq \cdots \subseteq R_n^*. \tag{20.2}$$

Since $K^* = K(\omega)$ is a pure extension, the radical tower (20.2) can be extended by adding $K \subseteq K^*$. □

§ 21. Group cohomology

Let G be a group and A be a **(left) G-module**. This means that A is an abelian group together with a map

$$G \times A \to A, \quad (g, a) \mapsto g \cdot a$$

such that $1 \cdot a = a$ for all $a \in A$, $(gh) \cdot a = g \cdot (h \cdot a)$ for all $g, h \in G$ and $a \in A$ and $g \cdot (a + b) = g \cdot a + g \cdot b$ for all $g \in G$ and $a, b \in A$.

21.1. Example. The group $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$ acts on $\mathbb{C}$ and $\mathbb{C}^\times$. Moreover, it acts trivially on $\mathbb{R}$ and $\mathbb{R}^\times$.

More generally, if E/K is a finite Galois extension, then the Galois group $\mathrm{Gal}(E/K)$ acts on E and $E^\times$.

21.2. Definition. Let G be a group and M and N be G-modules. A map $f : M \to N$ is a **homomorphism** of G-modules if $f(\sigma \cdot m) = \sigma \cdot f(m)$ for all $m \in M$ and $\sigma \in G$.

21.3. Definition. Let G be a group and M be a G-module. The submodule of **G-invariants** is defined as

$$M^G = \{m \in M : \sigma \cdot m = m \text{ for all } \sigma \in G\}.$$

Note that M^G is the largest submodule of the G-module M where G acts trivially. For example, if $G = \mathrm{Gal}(E/K)$, then $E^G = K$.

21.4. Proposition. *Let G be a group. If the sequence of G-modules and G-module homomorphism*

$$0 \longrightarrow P \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0$$

is exact, then

$$0 \longrightarrow P^G \xrightarrow{\alpha^0} M^G \xrightarrow{\beta^0} N^G$$

is exact, where α^0 is the restriction $\alpha|_{P^G}$ of α to P^G and β^0 is the restriction $\beta|_{M^G}$ of β to M^G.

Proof. Since α is injective, the restriction α^0 is injective.

Note that $\ker\beta^0 = \ker\beta \cap M^G \subseteq \ker\beta$.

We claim that $\alpha^0(P^G) = \alpha(P) \cap M^G$. If $m \in \alpha(P) \cap M^G$, then $\alpha(p) = m$ for some $p \in P$ and $\sigma \cdot m = m$. Since

$$\alpha(p) = m = \sigma \cdot m = \sigma \cdot \alpha(p) = \alpha(\sigma \cdot p),$$

$\sigma \cdot p - p \in \ker\alpha = \{0\}$. Hence $\sigma \cdot p = p$ and $p \in P^G$. Conversely, if $m \in \alpha^0(P^G)$, then $m = \alpha(p)$ for some $p \in P^G$. If $\sigma \in G$, then

$$\sigma \cdot m = \sigma \cdot \alpha(p) = \alpha(\sigma \cdot p) = \alpha(p) = m.$$

Hence $m \in M^G \cap \alpha(P)$.

Now

$$\alpha^0(P^G) = \alpha(P) \cap M^G = \ker\beta \cap M^G = \ker\beta^0. \qquad \square$$

Note that in the previous proposition, we did not prove that the map $\beta|_{M^G}$ is surjective.

21.5. Example. Let $G = \mathrm{Gal}(\mathbb{C}/\mathbb{R})$. Consider the following exact sequence of G-modules:

$$1 \longrightarrow \{-1, 1\} \longrightarrow \mathbb{C}^\times \xrightarrow{\beta} \mathbb{C}^\times \longrightarrow 1$$

where $\beta(z) = z^2$. Note that β is surjective. Take invariants to obtain the sequence

$$0 \longrightarrow \{-1, 1\} \longrightarrow \mathbb{R}^\times \xrightarrow{\beta^0} \mathbb{R}^\times$$

where $\beta^0(x) = x^2$. Note that β^0 is not surjective!

21.6. Definition. Let G be a group and N be a G-module. We define

$H^0(G, M) = M^G$,

$C^1(G, M) = \{\phi\colon G \to M : \phi \text{ is a map}\}$,

$Z^1(G, M) = \{\phi \in C^1(G, M) : \phi(\sigma\tau) = \phi(\sigma) + \sigma \cdot \phi(\tau) \text{ for all } \sigma, \tau \in G\}$,

Note that $Z^1(G, M)$ is an abelian group with the operation

$$(\phi + \phi_1)(\sigma) = \phi(\sigma) + \phi_1(\sigma).$$

Moreover, if $\phi \in Z^1(G, M)$, then $\phi(1_G) = 0_M$. To prove this fact, note that

$$\phi(1_G) = \phi(1_G 1_G) = \phi(1_G) + 1_G \cdot \phi(1_G) = \phi(1_G) + \phi(1_G)$$

implies that $\phi(1_G) = 0_M$.

21.7. Example. Let G be a group and M be a G-module. Fix $m \in M$. Then the map $\phi\colon G \to M$, $\phi(\sigma) = \sigma \cdot m - m$, is an element of $Z^1(G, M)$, because

$$\begin{aligned}
\phi(\sigma\tau) &= (\sigma\tau) \cdot m - m \\
&= (\sigma\tau) \cdot m - \sigma \cdot m + \sigma \cdot m - m \\
&= \sigma \cdot (\tau \cdot m - m) + \sigma \cdot m - m \\
&= \sigma \cdot \phi(\tau) + \phi(\sigma)
\end{aligned}$$

for all $\sigma, \tau \in G$.

21.8. Definition. Let G be a group and M be a G-module. The set $B^1(G, M)$ of **coboundaries** is the set of elements $\phi \in C^1(G, M)$ such that there is a fixed $m \in M$ such that $\phi(\sigma) = \sigma \cdot m = m$ for all $\sigma \in G$.

We proved in Example 21.7 that $B^1(G, M) \subseteq Z^1(G, M)$. A direct calculation shows that, in fact, $B^1(G, M)$ is a subgroup of $Z^1(G, M)$.

21.9. Definition. Let G be a group and M be a G-module. The **first cohomology group** of G with coefficients in M is defined as the quotient

$$H^1(G, M) = Z^1(G, M)/B^1(G, M).$$

21.10. Example. If G acts trivially on M, then

$$H^0(G, M) = M^G = M, \quad B^1(G, M) = \{0\}, \quad Z^1(G, M) = \operatorname{Hom}(G, M).$$

Hence $H^1(G, M) \simeq \operatorname{Hom}(G, M)$.

21.11. Example. Let $G = \operatorname{Gal}(\mathbb{C}/\mathbb{R}) = \{\mathrm{id}, \gamma\}$, where $\gamma\colon \mathbb{C} \to \mathbb{C}$, $z \mapsto \overline{z}$, is the complex conjugation. Then

$$H^0(G, \mathbb{R}^\times) = \left(\mathbb{R}^\times\right)^G = \mathbb{R}^\times.$$

Since G acts trivially on $\mathbb{R}^\times$,

$$H^1(G, \mathbb{R}^\times) = \operatorname{Hom}(G, \mathbb{R}^\times) \simeq \operatorname{Hom}(G, \{-1, 1\}) \simeq \mathbb{Z}/2.$$

The following lemma will be useful.

21.12. Lemma. *Let G be a group and $\alpha\colon M \to N$ be a homomorphism of G-modules. Then*

$$\alpha^1\colon H^1(G, M) \to H^1(G, N), \quad \phi + B^1(G, M) \mapsto \alpha \circ \phi + B^1(G, N),$$

is a group homomorphism.

Proof. Let us prove that the map α^1 is well-defined. If $\phi - \phi' \in B^1(G, M)$, then there exists a fixed $m \in M$ such that $(\phi - \phi')(\sigma) = \sigma \cdot m - m$ for all $\sigma \in G$. Let $n = \alpha(m) \in N$. For $\sigma \in G$,

$$\alpha((\phi - \phi')(\sigma)) = \alpha(\sigma \cdot m - m) = \sigma \cdot \alpha(m) - \alpha(m) = \sigma \cdot n - n.$$

Thus $\alpha \circ \phi - \alpha \circ \phi' \in B^1(G, N)$.

We now prove that α^1 is a group homomorphism. If $\phi,\phi'\in Z^1(G,M)$, then

$$\begin{aligned}\alpha^1(\phi+B^1(G,M)+\phi'+B^1(G,M)) &= \alpha^1(\phi+\phi'+B^1(G,M))\\ &= \alpha\circ(\phi+\phi')+B^1(G,N)\\ &= \alpha\circ\phi+\alpha\circ\phi'+B^1(G,N)\\ &= \alpha\circ\phi+B^1(G,N)+\alpha\circ\phi'+B^1(G,N)\\ &= \alpha^1(\phi+B^1(G,M))+\alpha^1(\phi'+B^1(G,M)). \qquad\square\end{aligned}$$

We will provide a detailed proof of the upcoming result. The theorem will be established by applying a **diagram chasing** technique, a widely used tool in homological algebra. The proof is tedious and may seem intricate, but the method essentially involves "chasing" elements around a (commutative) diagram until we attain the desired result.

21.13. Theorem. *Let G be a group and*

$$0 \longrightarrow P \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0$$

be an exact sequence of G-modules and G-module homomorphism. Then there exists a ***connection homomorphism*** *δ such that the sequence*

$$\begin{tikzcd} 0 \arrow[r] & H^0(G,P) \arrow[r, "\alpha^0"] & H^0(G,M) \arrow[r, "\beta^0"] & H^0(G,N) \arrow[dll, "\delta"] \\ & H^1(G,P) \arrow[r, "\alpha^1"] & H^1(G,M) \arrow[r, "\beta^1"] & H^1(G,N) \end{tikzcd} \tag{21.1}$$

of abelian groups and group homomorphisms is exact.

Proof. By Proposition 21.4, the long sequence (21.1) is exact at $H^0(G,P)=P^G$, $H^0(G,M)=M^G$ and $H^0(G,N)=N^G$. Note that, in particular, $\alpha\colon P\to\alpha(P)$ is a bijective group homomorphism.

Let us construct the connecting homomorphism $\delta\colon H^0(G,N)\to H^1(G,P)$. For $n\in N^G$, let $m\in M$ be such that $\beta(m)=n$. We define $\delta(n)=\phi+B^1(G,P)$, where

$$\phi(\sigma)=\alpha^{-1}(\sigma\cdot m-m).$$

Note that $\sigma\cdot m-m\in\operatorname{im}\alpha=\ker\beta$, as

$$\beta(\sigma\cdot m-m)=\sigma\cdot\beta(m)-\beta(m)=\sigma\cdot n-n=0.$$

Let us prove that the map δ is well-defined: if $m,m'\in M$ are such that

$$\beta(m)=\beta(m')=n,$$

then $m-m'\in\ker\beta=\alpha(P)$. For $\sigma\in G$, write $\phi'(\sigma)=\sigma\cdot m'-m'$. Since $m-m'=\alpha(p)$ for some $p\in P$ and α^{-1} is a homomorphism of G-modules,

$$\begin{aligned}\phi(\sigma)-\phi'(\sigma) &= \alpha^{-1}(\sigma\cdot m-m)-\alpha^{-1}(\sigma\cdot m'-m')\\ &= \alpha^{-1}(\sigma\cdot(m-m'))-\alpha^{-1}(m-m')\\ &= \alpha^{-1}(\sigma\cdot\alpha(p)-\alpha(p))\\ &= \sigma\cdot p-p.\end{aligned}$$

Thus $\phi-\phi'\in B^1(G,P)$.

Note that $\phi \in Z^1(G,P)$, because

$$\begin{aligned}\phi(\sigma\tau) &= \alpha^{-1}((\sigma\tau)\cdot m - m)\\ &= \alpha^{-1}((\sigma\tau)\cdot m - \sigma\cdot m + \sigma\cdot m - m)\\ &= \alpha^{-1}(\sigma\cdot(\tau\cdot m - m)) + \alpha^{-1}(\sigma\cdot m - m)\\ &= \sigma\cdot\phi(\tau) + \phi(\sigma)\end{aligned}$$

holds for all $\sigma,\tau \in G$.

We now prove that the sequence (21.1) is exact at $H^0(G,N) = N^G$. We need to prove that $\ker\delta = \operatorname{im}\beta^0$. To prove $\supseteq$ note that if $m \in M^G$ is such that $\delta(\beta(m)) = \phi + B^1(G,P)$, then

$$\phi(\sigma) = \alpha^{-1}(\sigma\cdot m - m) = 0.$$

Conversely, if $n \in \ker\delta$, then there exists $m \in M$ such that $\beta(m) = n$ and

$$\delta(\beta(m)) = \phi + B^1(G,P),$$

where $\phi \in B^1(G,P)$ and $\phi(\sigma) = \alpha^{-1}(\sigma\cdot m - m)$ for all $\sigma \in G$. Since $\phi \in B^1(G,P)$, there exists $p \in P$ such that $\phi(\sigma) = \sigma\cdot p - p$ for all $\sigma \in G$. Note that

$$\beta(m - \alpha(p)) = \beta(m) - \beta(\alpha(p)) = \beta(m) = n.$$

Moreover, $m - \alpha(p) \in M^G$, as $\sigma\cdot(m - \alpha(p)) = m - \alpha(p)$. Hence $n \in \operatorname{im}\beta^0$.

We now prove that (21.1) is exact at $H^1(G,P)$, that is $\operatorname{im}\delta = \ker\alpha^1$. To prove $\subseteq$ note that for $n \in N^G$, $\delta(n) = \phi + B^1(G,P)$, where $\phi(\sigma) = \alpha^{-1}(\sigma\cdot m - m)$ for all $\sigma \in G$ and some $m \in M$ such that $\beta(m) = n$. In particular, $\alpha\circ\phi \in B^1(G,M)$. Then

$$\alpha^1(\phi + B^1(G,P)) = \alpha\circ\phi + B^1(G,M) = B^1(G,M).$$

To prove $\supseteq$, let $\phi + B^1(G,P) \in \ker\alpha^1$. Then $\alpha\circ\phi \in B^1(G,M)$, that is $\alpha(\phi(\sigma)) = \sigma\cdot m - m$ for all $\sigma \in G$ and some $m \in M$. Note that

$$\delta(\beta(m)) = \psi + B^1(G,P),$$

where $\psi(\sigma) = \alpha^{-1}(\sigma\cdot m - m)$. This implies that $\alpha(\psi(\sigma)) = \alpha(\phi(\sigma))$ for all $\sigma \in G$. Since α is injective, $\psi = \phi$. Therefore $\phi + B^1(G,P)$ belongs to the image of δ.

Finally, we prove that the sequence (21.1) is exact at $H^1(G,M)$, that is $\operatorname{im}\alpha^1 = \ker\beta^1$. To prove $\subseteq$ note that

$$\beta^1(\alpha^1(\phi + B^1(G,P))) = \beta^1(\alpha\circ\phi + B^1(G,M)) = (\beta\circ\alpha)\circ\phi + B^1(G,N) = B^1(G,N).$$

Conversely, let $\phi + B^1(G,M) \in \ker\beta_1$. Then $\beta\circ\phi \in B^1(G,N)$. Thus there exists $n \in N$ such that $\beta(\phi(\sigma)) = \sigma\cdot n - n$ for all $\sigma \in G$. Since β is surjective, $n = \beta(m)$ for some $m \in M$. Hence

$$\beta(\phi(\sigma)) = \sigma\cdot n - n = \sigma\cdot\beta(m) - \beta(m) = \beta(\sigma\cdot m - m).$$

For each $\sigma \in G$, $\phi(\sigma) - (\sigma\cdot m - m) \in \ker\beta = \operatorname{im}\alpha$. and therefore $\phi(\sigma) - (\sigma\cdot m - m) = \alpha(\rho_\sigma)$. This defines a map $\rho\colon G \to P$, $\sigma \mapsto \rho_\sigma$. We claim that $\rho \in Z^1(G,P)$. If $\sigma,\tau \in G$, then

$$\begin{aligned}\alpha(\rho_{\sigma\tau}) &= \phi(\sigma\tau) - ((\sigma\tau)\cdot m - m)\\ &= \phi(\sigma) + \sigma\cdot\phi(\tau) - (\sigma\cdot(\tau\cdot m - m) + \sigma\cdot m - m)\\ &= \alpha(\rho_\sigma) + \sigma\cdot\alpha(\rho_\tau).\end{aligned}$$

Since α is injective, it follows that $\rho \in Z^1(G,P)$. Now

$$\alpha_1(\rho + B^1(G,P)) = \alpha\circ\rho + B^1(G,M) = \phi + B^1(G,M).$$ □

21.14. Theorem. *Let G be a finite group and M be a G-module. Then*

$$|G|H^1(G,M)=\{0\}.$$

Proof. Let $n=|G|$. It is enough to prove that if $\phi\in Z^1(G,M)$, then $n\phi\in B^1(G,M)$. Let $\phi\in Z^1(G,M)$ and $\sigma\in G$. Then

$$\phi(\sigma\tau)=\phi(\sigma)+\sigma\cdot\phi(\tau)$$

for all $\tau\in G$. Summing over all possible $\tau\in G$, we obtain that

$$\sum_{\tau\in G}\phi(\tau)=\sum_{\tau\in G}\phi(\sigma\tau)=\sum_{\tau\in G}\sigma\cdot\phi(\tau)+\sum_{\tau\in G}\phi(\sigma)=n\phi(\sigma). \tag{21.2}$$

Let $m=-\sum_{\tau\in G}\phi(\tau)\in M$. Then (21.2) can be rewritten as

$$-m=\sum_{\tau\in G}\phi(\tau)=\sigma\cdot\sum_{\tau\in G}\phi(\tau)+n\phi(\sigma)=-\sigma\cdot m+n\phi(\sigma).$$

Thus $n\phi(\sigma)=\sigma\cdot m-m$ and hence $n\phi\in B^1(G,M)$. □

21.15. Exercise. Let G be a finite group and M be a finite G-module of size coprime to $|G|$. Prove that $H^1(G,M)=\{0\}$.

21.16. Exercise. Let G be a finite group and M be a finitely generated G-module. Prove that $H^1(G,M)$ is finite.

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